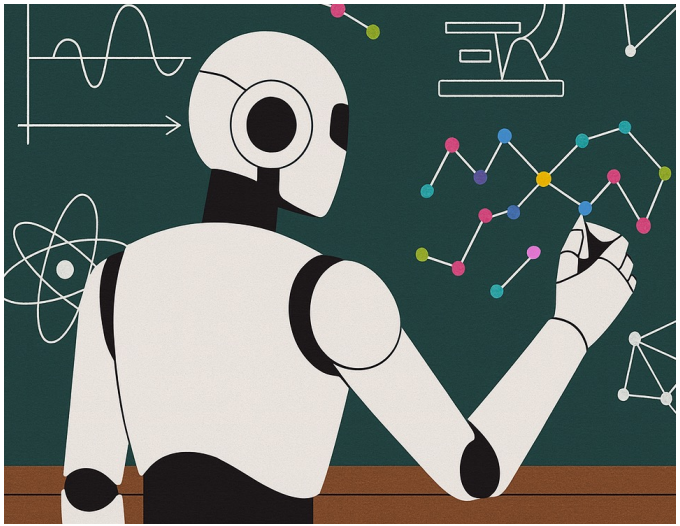
AI for (Quantum) Physics

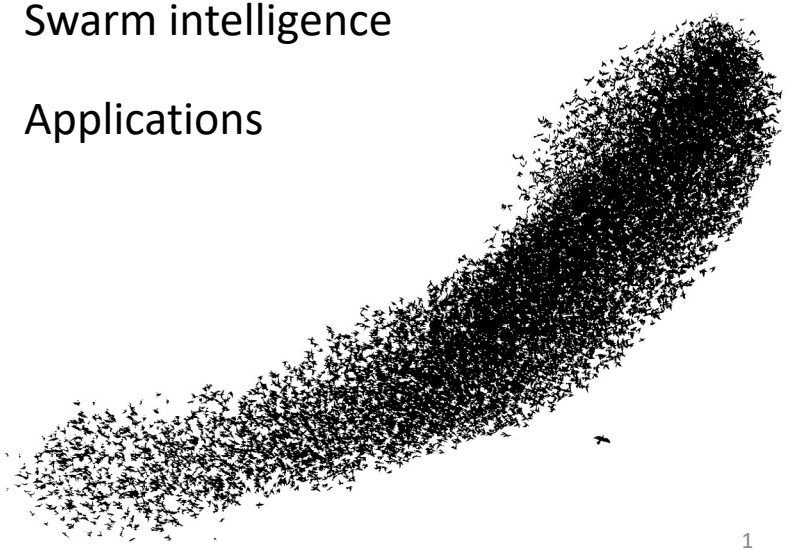


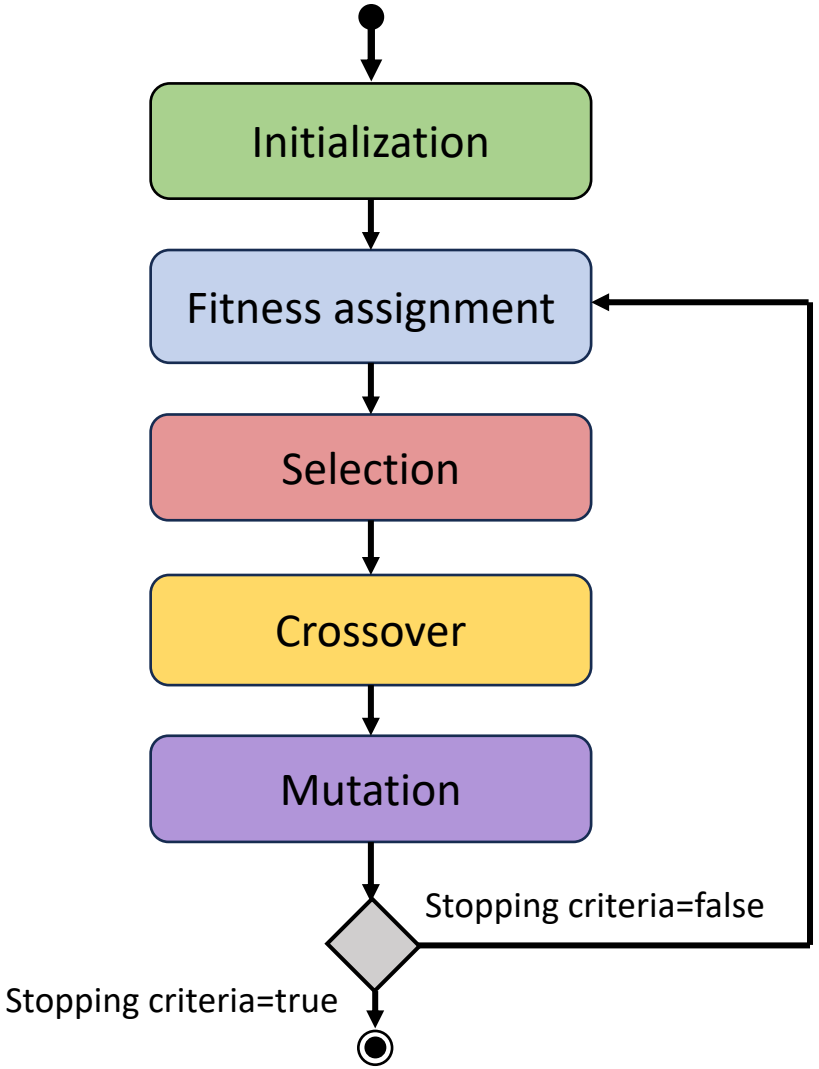
Xuemei Gu
xuemei.gu@uni-jena.de

If you have any questions, feel free to reach out!

Today's topic:

- Swarm intelligence
- Applications




Initialization
Fitness assignment
Selection
Crossover
Mutation
Stopping criteria=false
Stopping criteria=true



Collective and Swarm Intelligence

Nature’s inspiration, general principles, self-organization

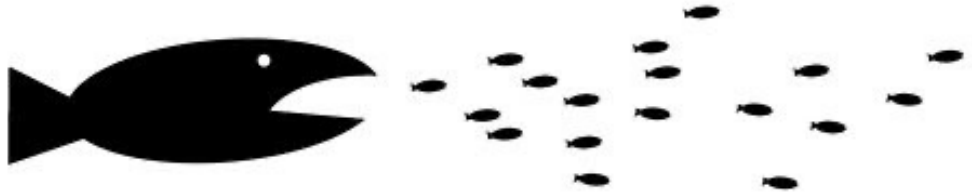
acting as a unit

Many is better than one (?)

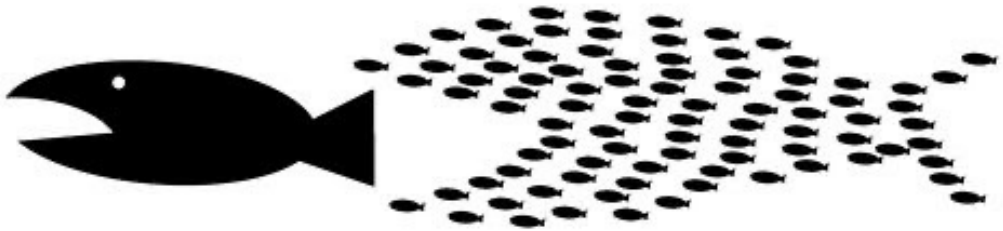


https://www.youtube.com/watch?v=V96hXqw2oU8

Swarm intelligence Manifesto



Don’t Panic,



Organize!

Fish schooling



Birds flocking



Ant Collaboration





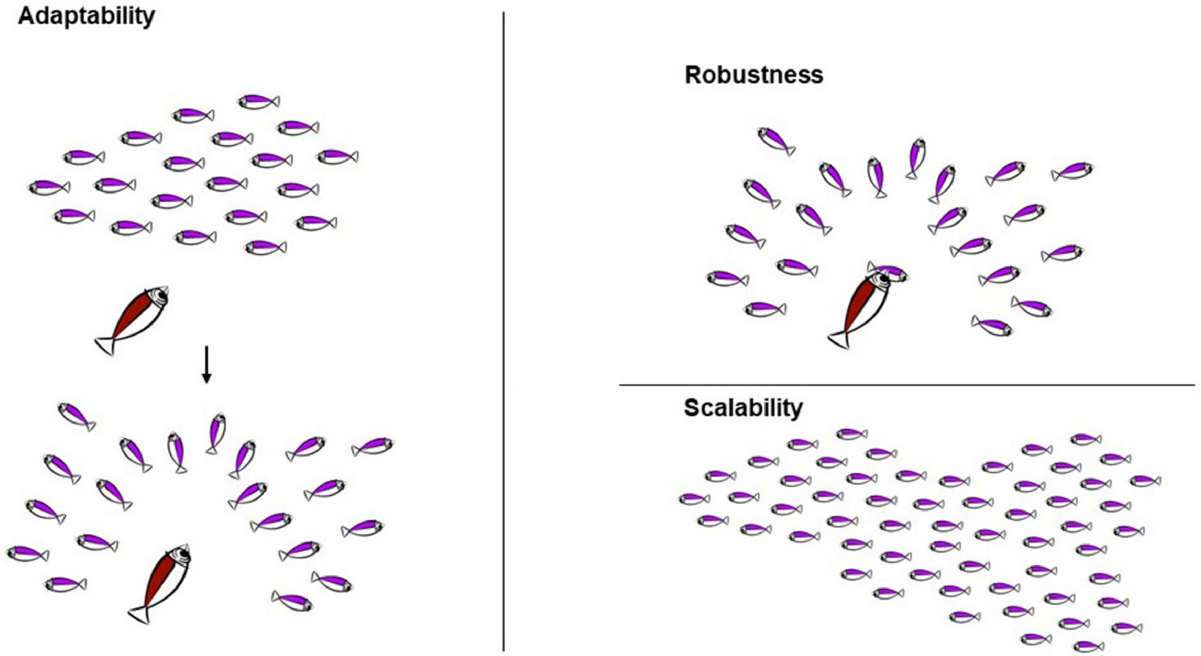
Termites' nest

What do all these behaviours have in common?

- Distributed system of simple, autonomous individuals
- No central controller — coordination is fully decentralized
- Interactions are purely local (e.g., neighbours, pheromones, vibrations)
- Agents follow simple behavioural rules
- Global patterns emerge that exceed the ability of any single individual
- Cooperation arises through self-organization, not planning
- Information spreads gradually through neighbour-to-neighbour transmission

What do all these behaviours have in common?

Image from Schranz, Melanie, et al. Swarm and Evolutionary Computation 60 (2021): 100762.



- **Adaptivity** (rapid response to environmental change)
- **Robustness** (failure of individuals does not break the system)
- **Scalability** (performance often improves as the group grows)

Swarm Intelligence

Swarm intelligence is the collective behaviour of decentralized, self-organized system, natural or artificial.

Swarm Intelligence design applied the same principles to obtain the same objective as in Nature's complex adaptive systems. It is an emerging field of biologically-inspired artificial intelligence based on the self organized collective behaviour of social swarms in nature.





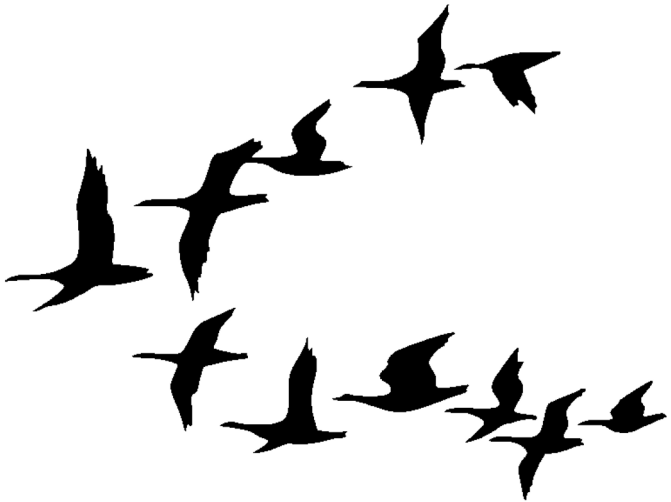


The expression was introduced by Gerardo Beni and Jing Wang in 1989, in the context of cellular robotic systems

The most popular algorithms in the swarm intelligence domain

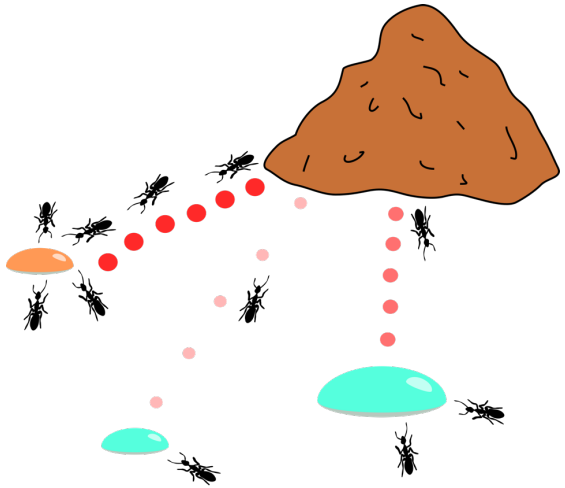
Particle Swarm Optimization (PSO)

PSO was proposed by Kennedy and Eberhart in 1995, inspired by the social behaviour of bird flocking or fish schooling



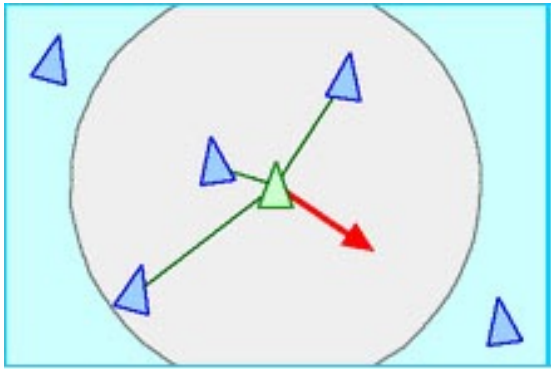
Ant Colonies Optimization (ACO)

ACO is a probabilistic technique for solving computation problems which can be reduced to finding good paths through graphs



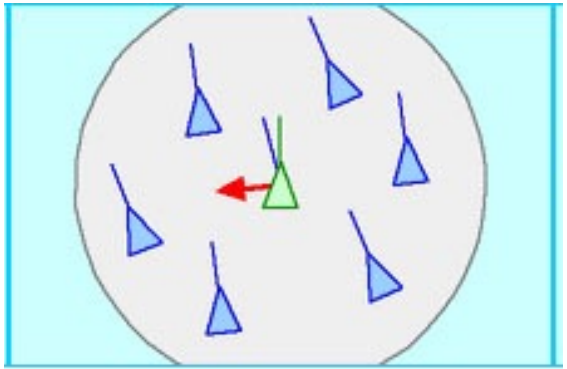
Particle Swarm Optimization (PSO)

In 1986, Craig Reynolds described this process in 3 very behaviours, which simulates the flocking behaviour of birds and related group motion.



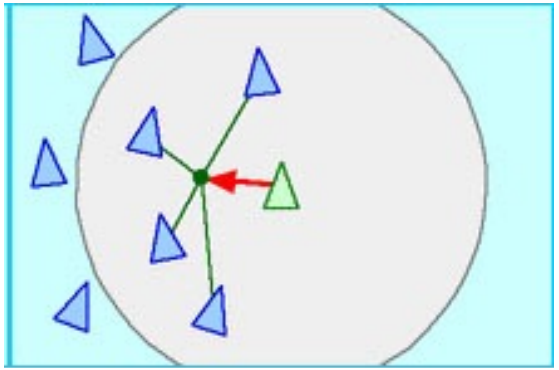
Separation

Avoid crowding local flock mates



Alignment

Move towards the average heading of local flock mates



Cohesion

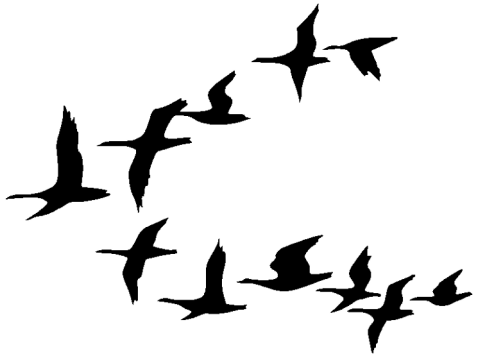
Move towards the average position of local flock mates

Particle Swarm Optimization (PSO)

proposed by Kennedy and Eberhart in 1995

Uses a number of agents (particles) that constitute a swarm moving around in the search space looking for the best solution

Each particle in the search space adjusts its “flying” according to its own flying experience as well as the flying experience of its colleagues.



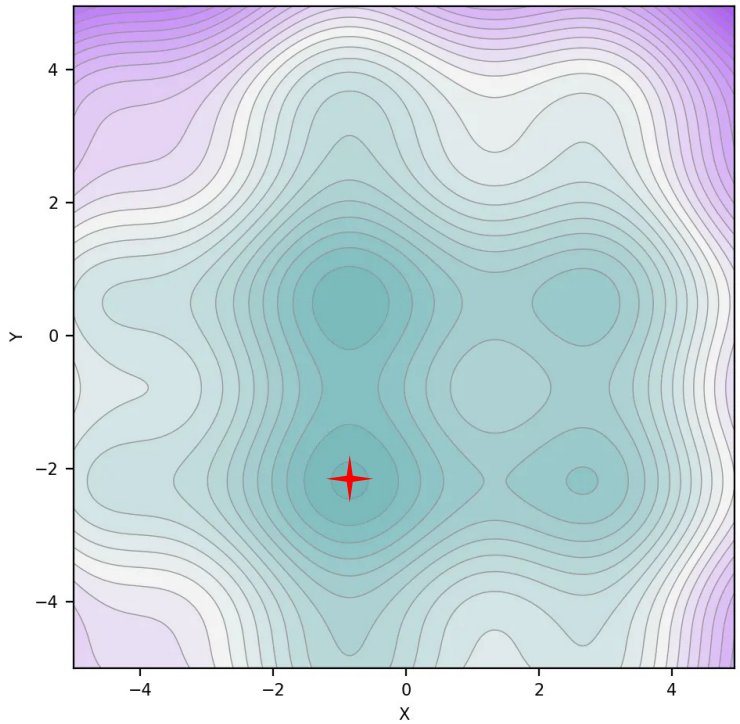
Collection of flying particles (swarm) - Changing solutions

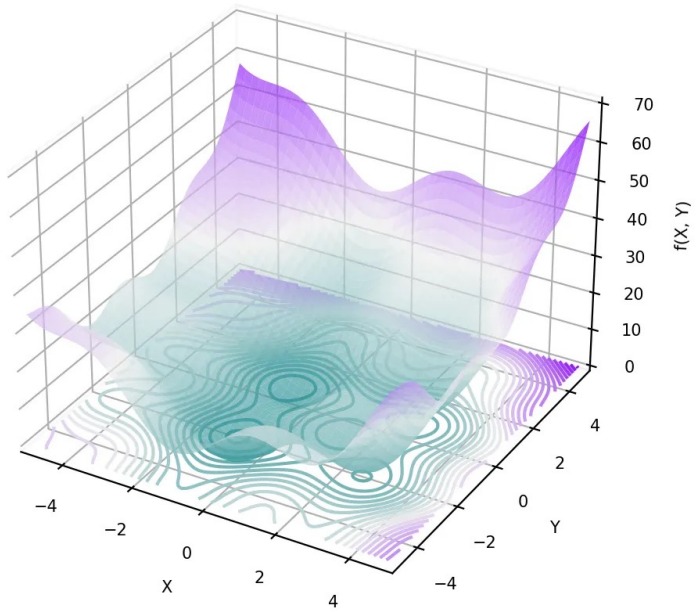
Search area - possible solutions

Movement towards a promising area to get the global optimum

Particle Swarm Optimization (PSO)

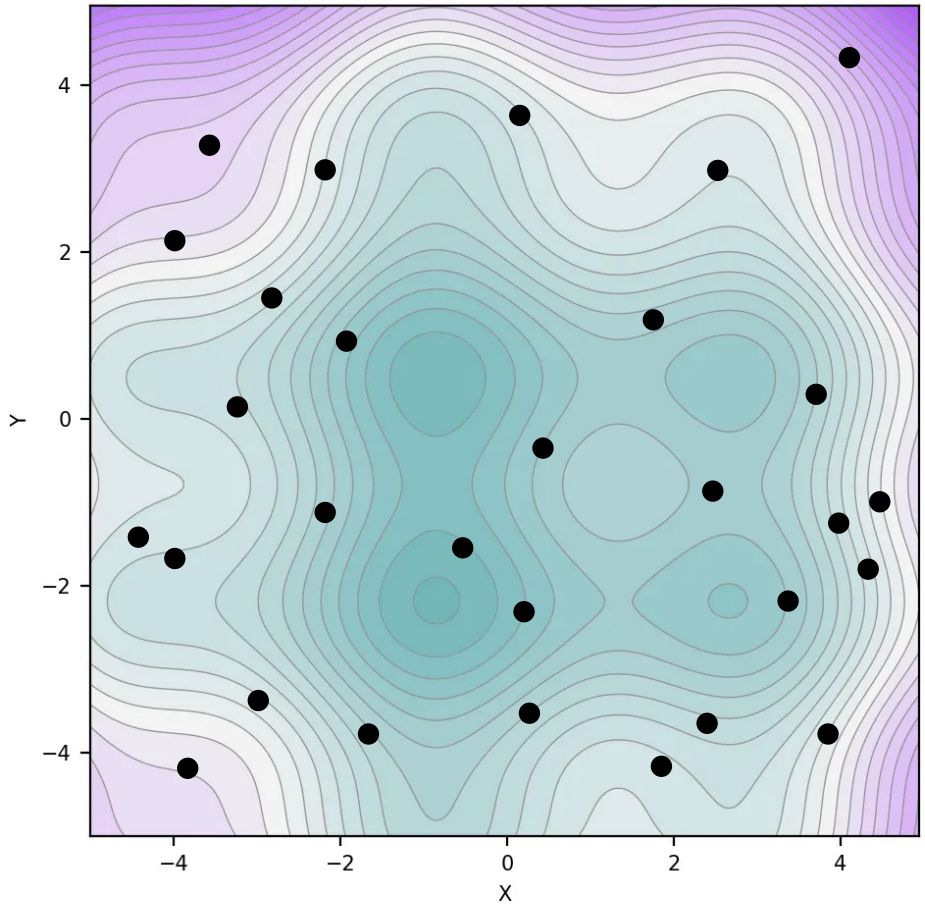
$$f(x,y) = x^2 + (y+1)^2 - 5cos(\frac{3}{2}x + \frac{3}{2}) - 3cos(2y - \frac{3}{2})$$



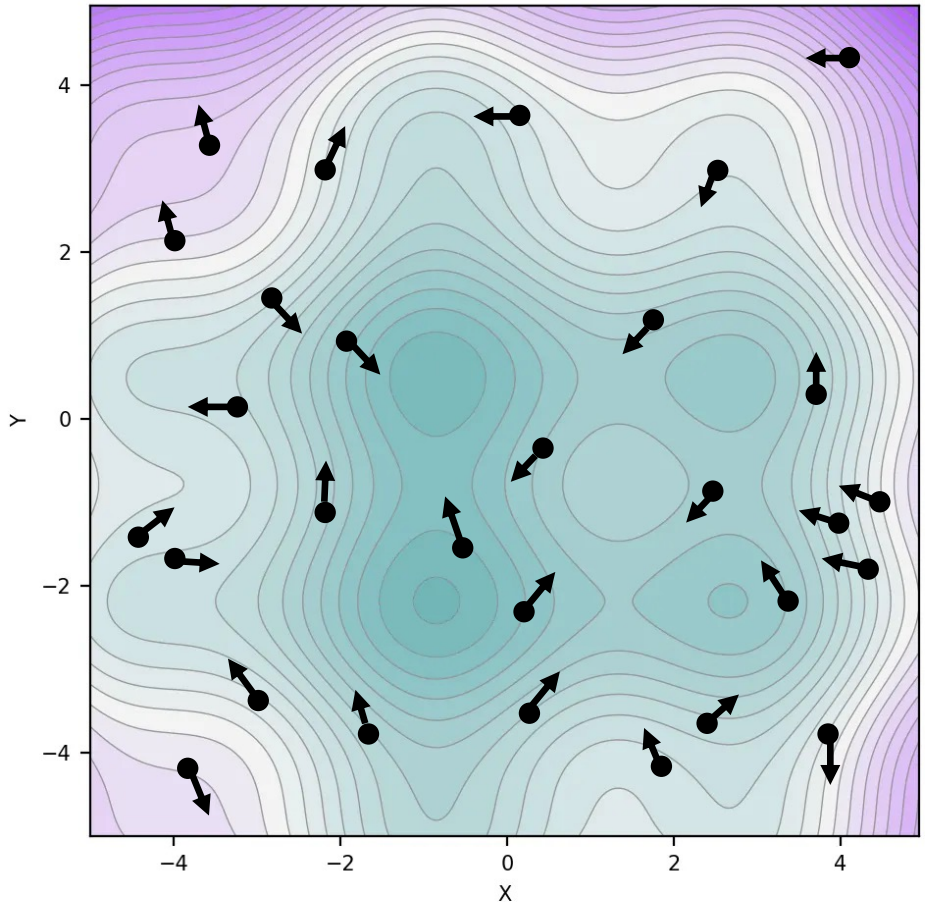


from Axel Theveno's blog

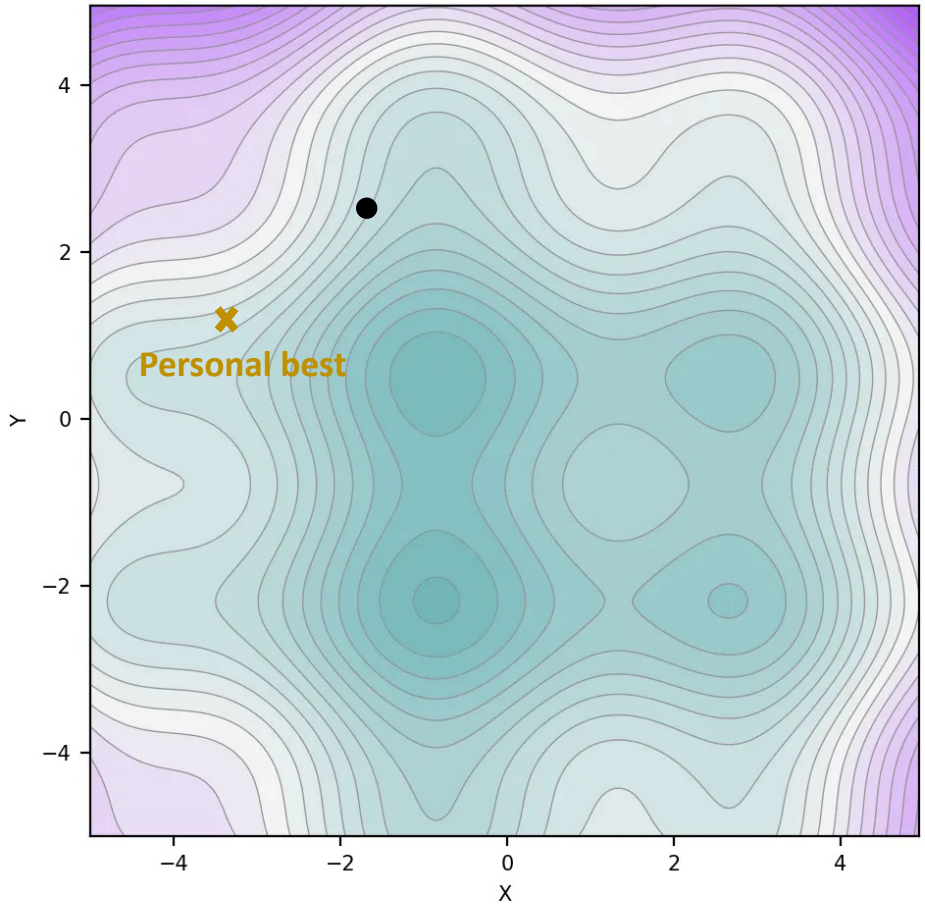
Particle Swarm Optimization (PSO)



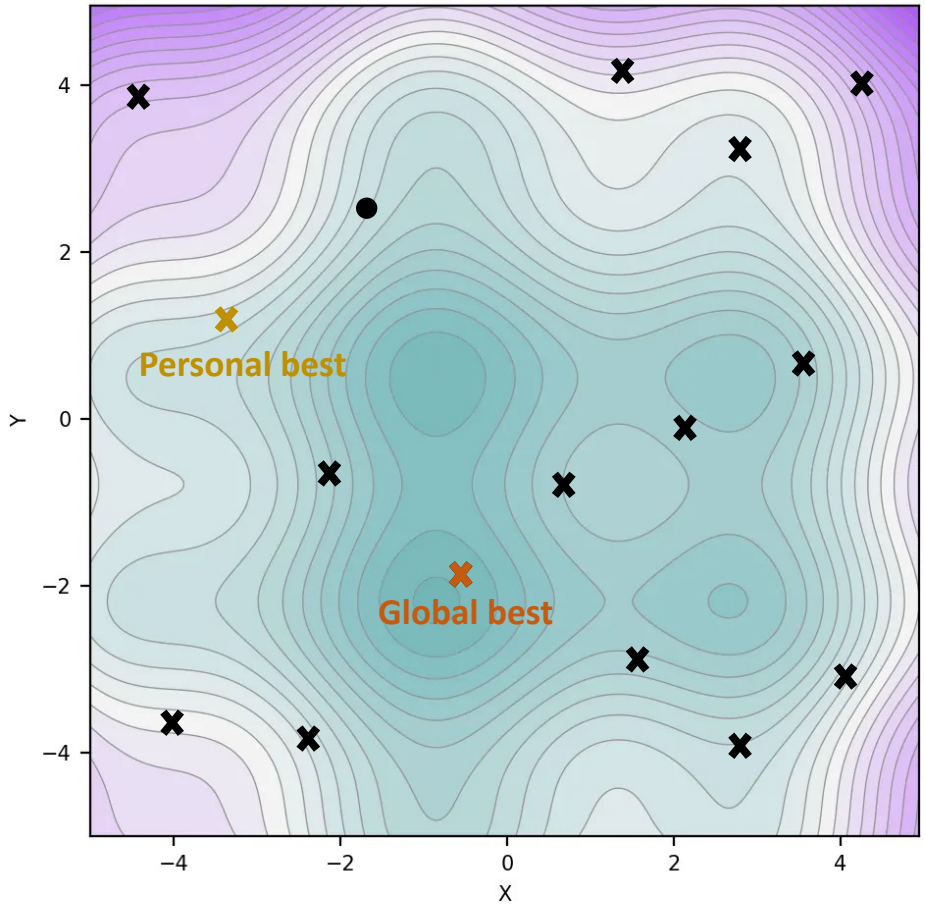
Particle Swarm Optimization (PSO)



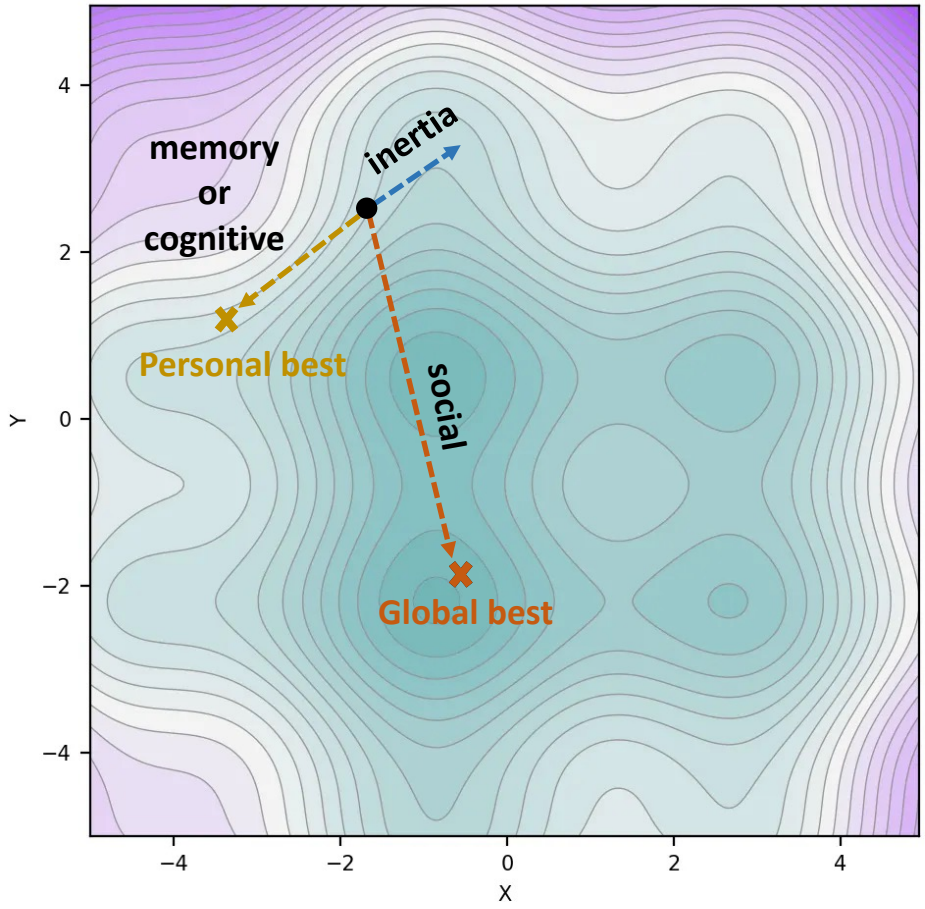
Particle Swarm Optimization (PSO)



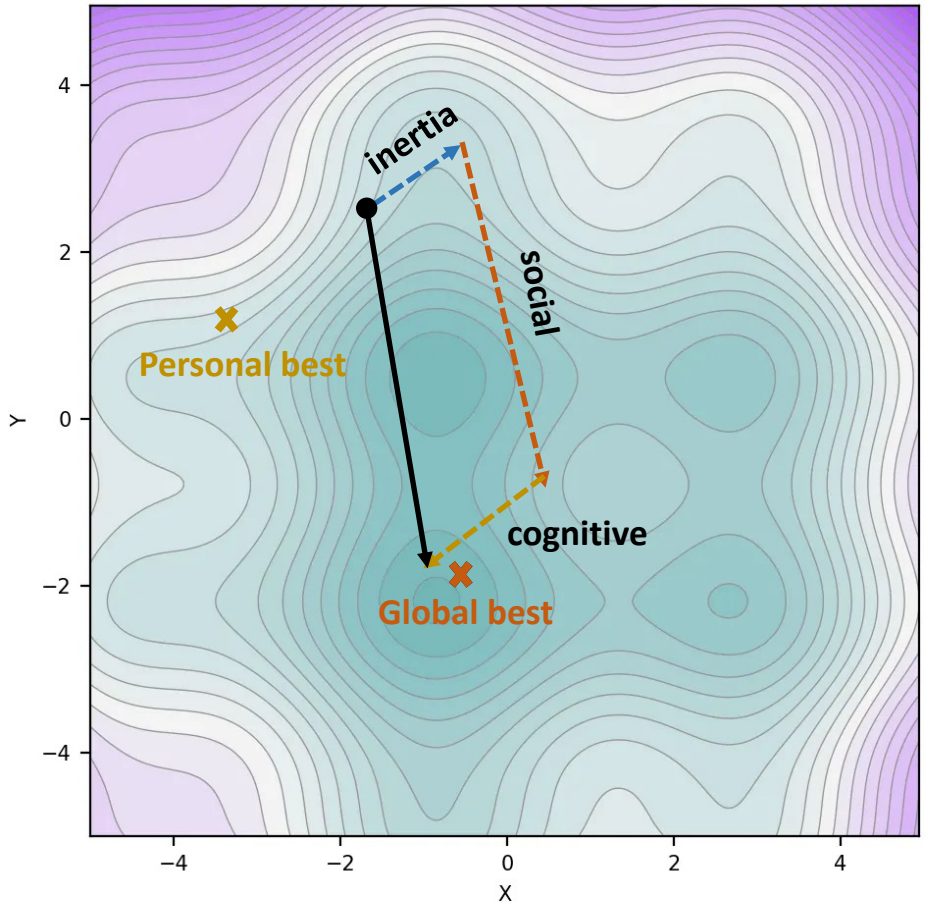
Particle Swarm Optimization (PSO)



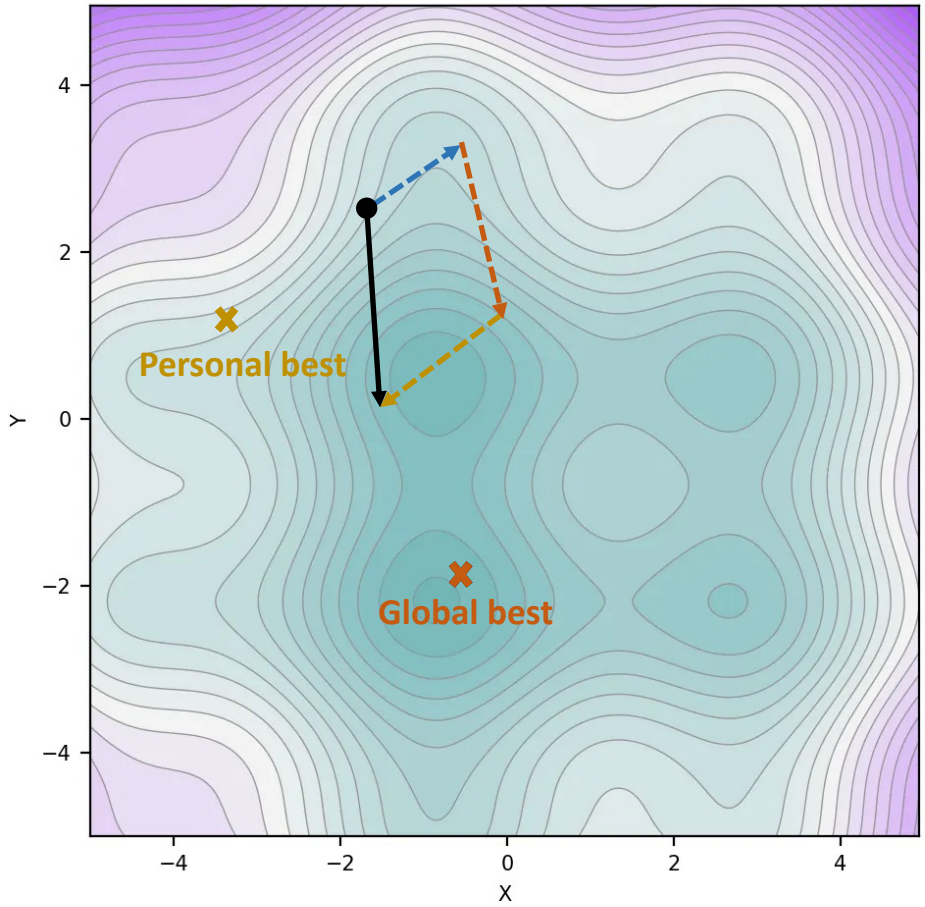
Particle Swarm Optimization (PSO)



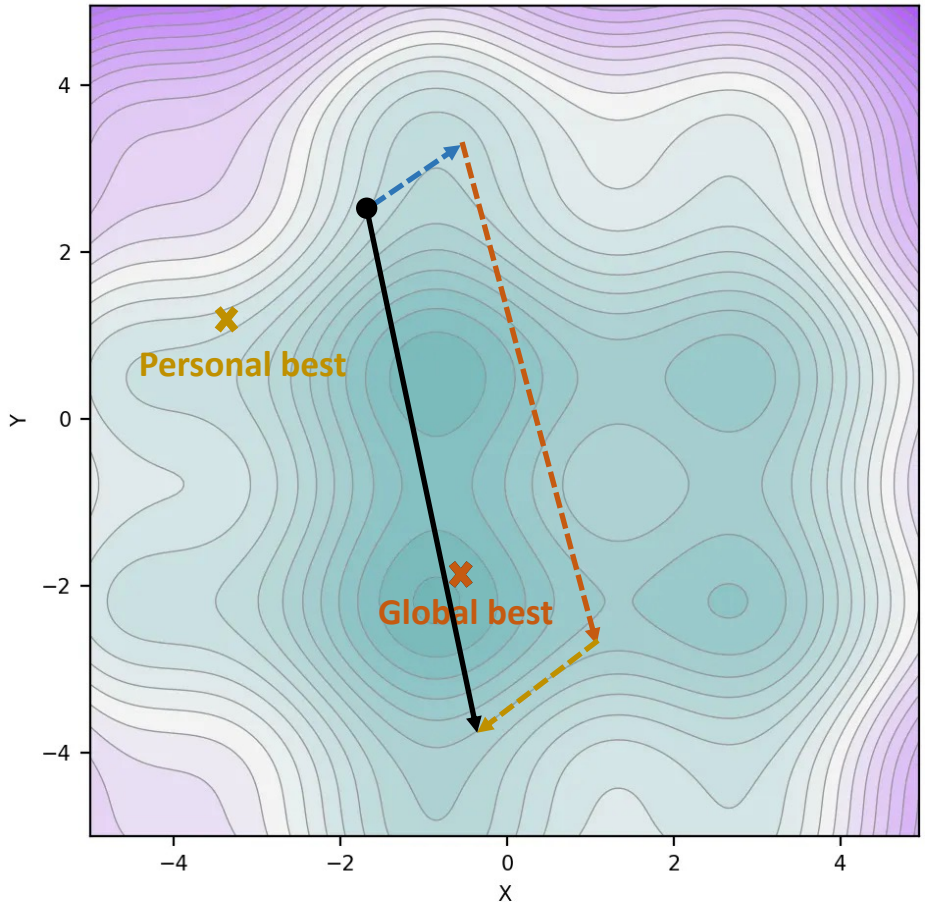
Particle Swarm Optimization (PSO)



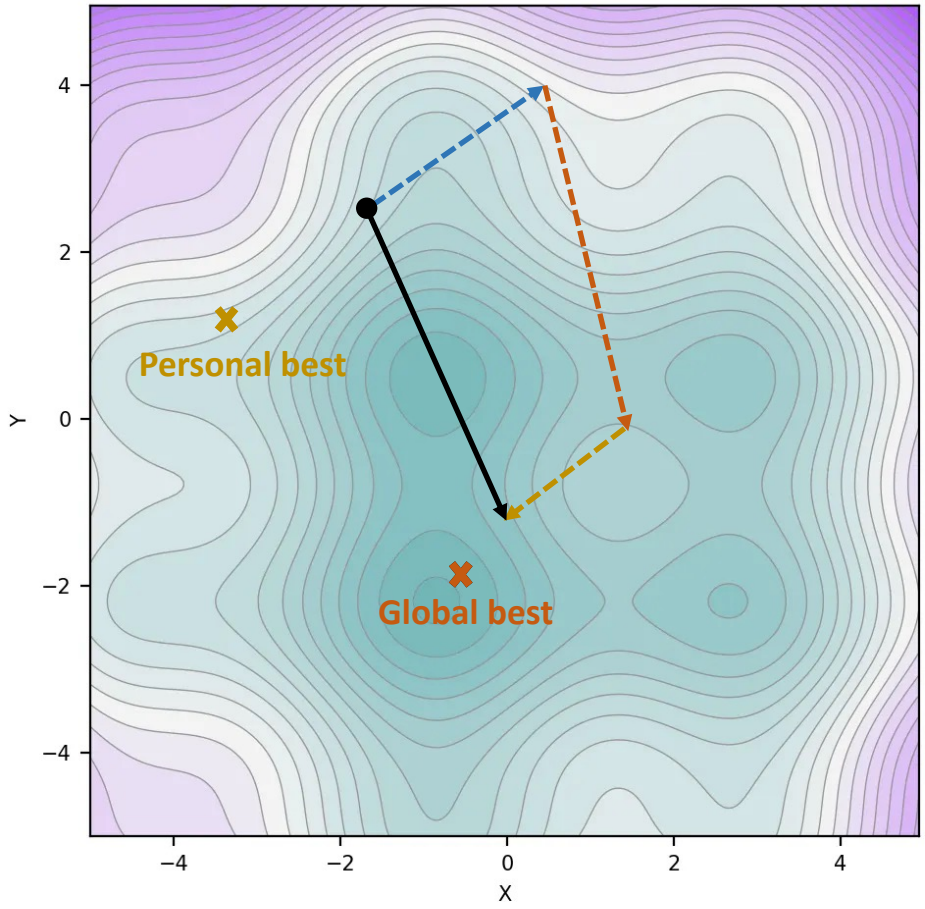
Particle Swarm Optimization (PSO)



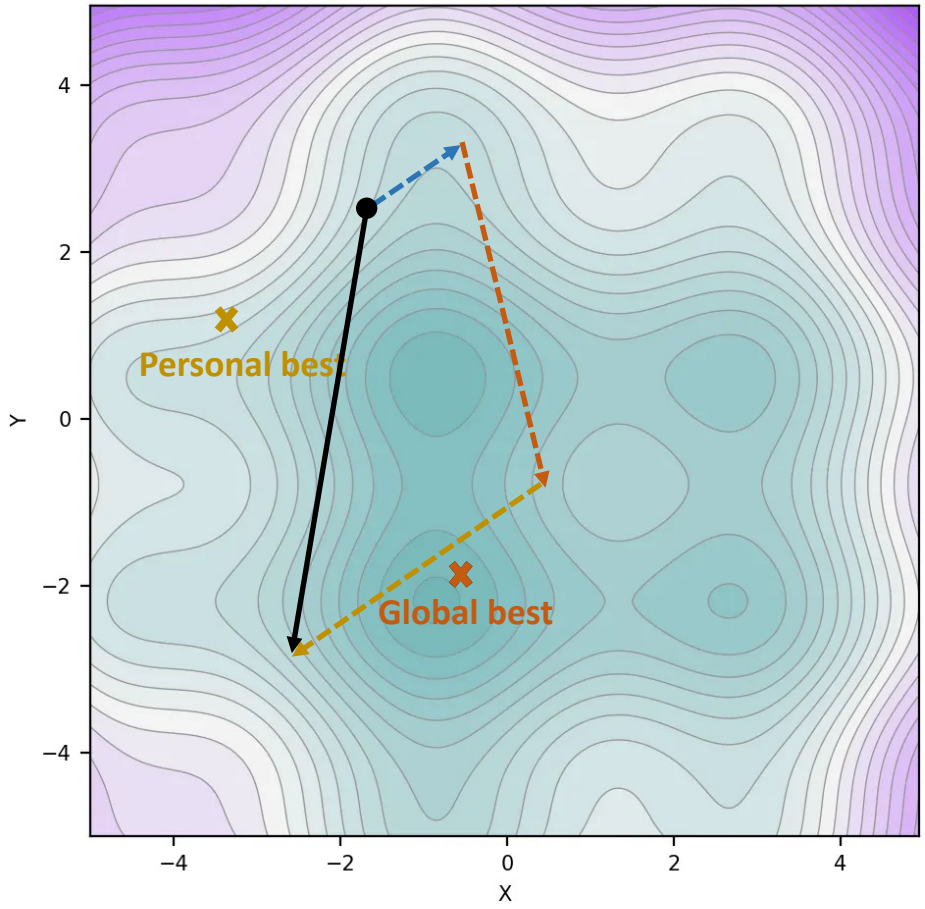
Particle Swarm Optimization (PSO)



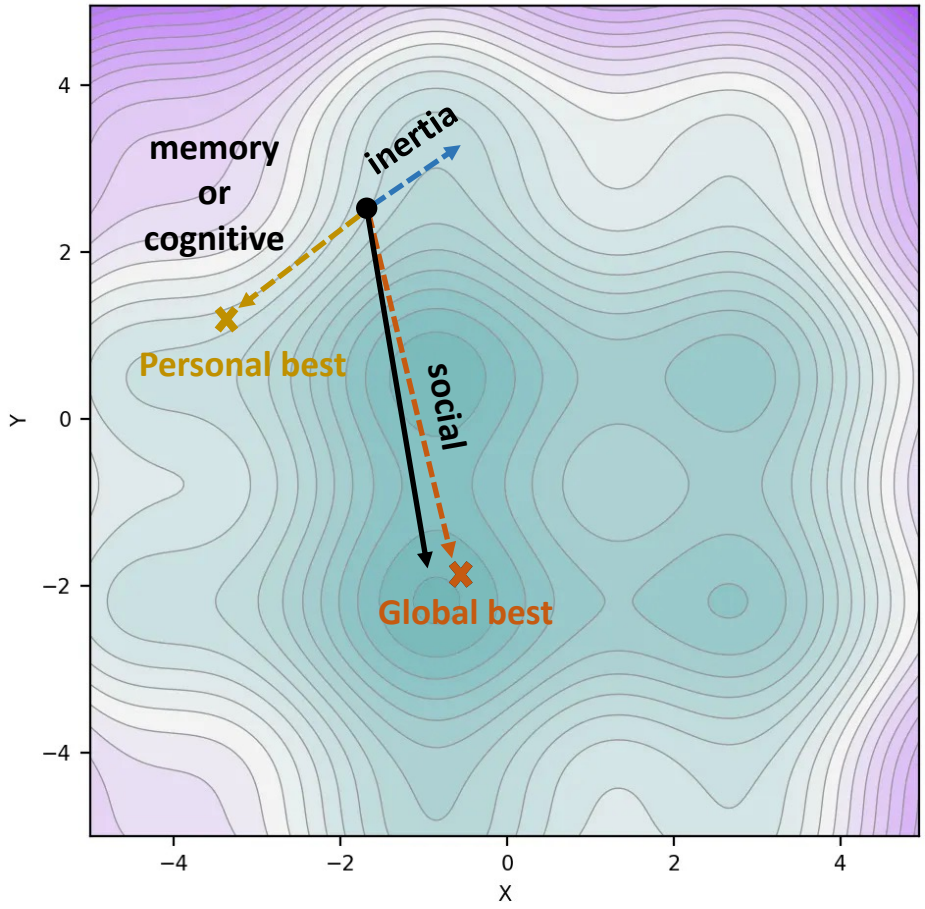
Particle Swarm Optimization (PSO)



Particle Swarm Optimization (PSO)



Particle Swarm Optimization (PSO)



Particle Swarm Optimization (PSO)

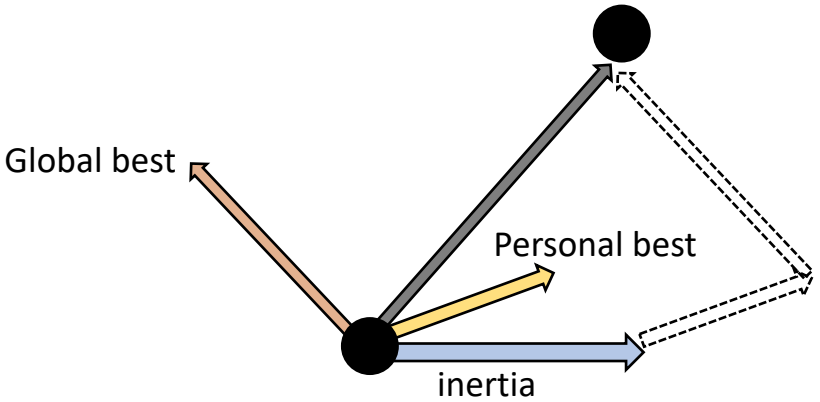
Update rules:

r_1, r_2 random values in [0,1]

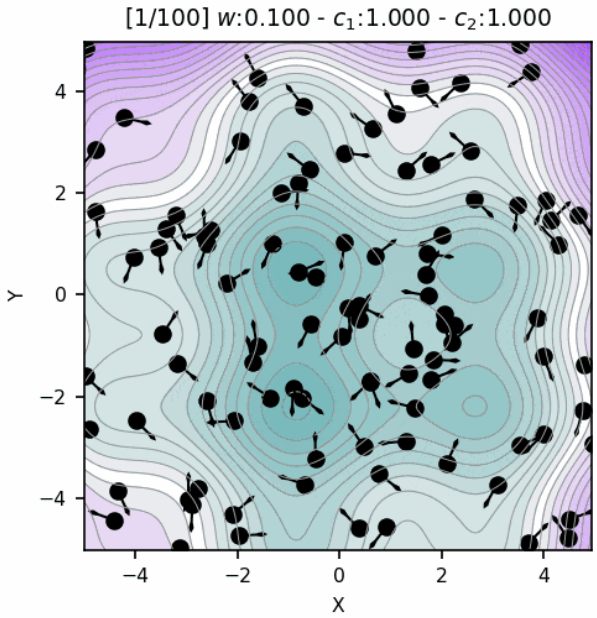
$$x_i^{t+1} = x_i^t + v_i^{t+1}$$

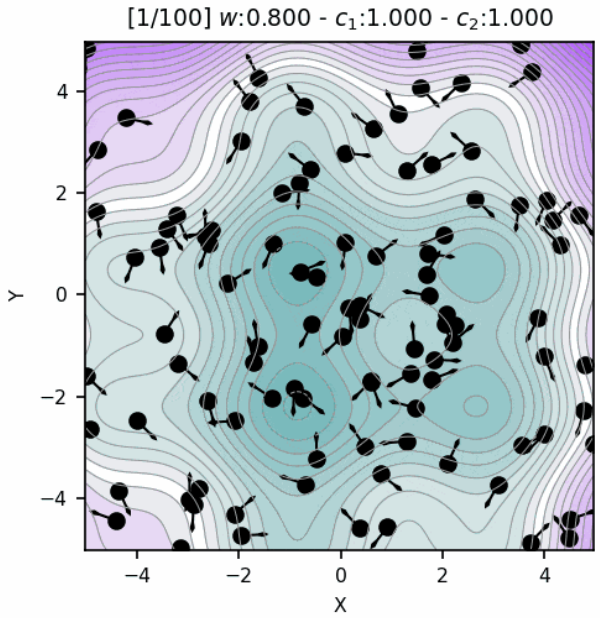
$$v_i^{t+1} = \omega v_i^t + c_1 r_1 (x_{\text{best}(i)}^t - x_i^t) + c_2 r_2 (x_{\text{global}}^t - x_i^t)$$

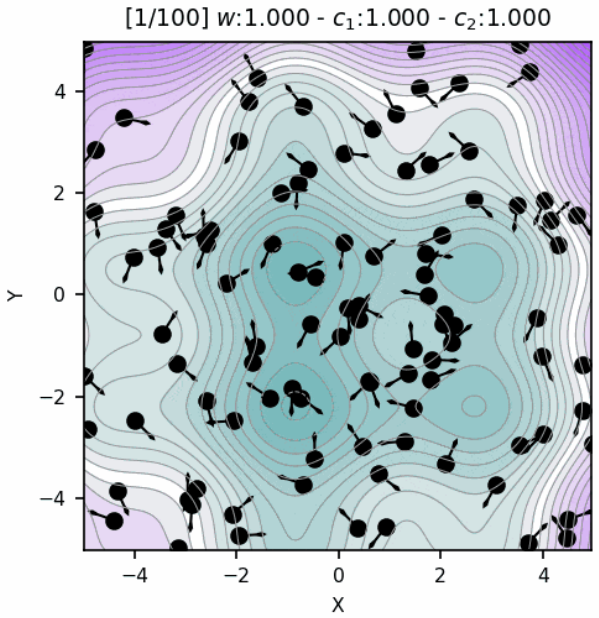
Inertia **Cognitive (Personal)** **Social (Global)**



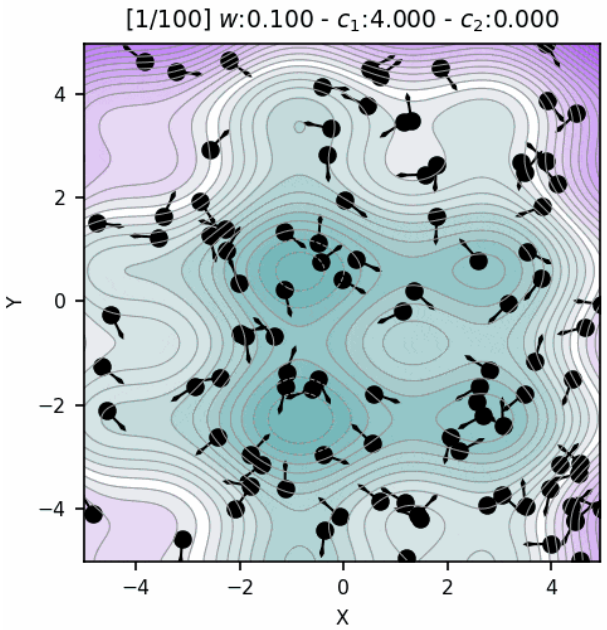
Particle Swarm Optimization (PSO)

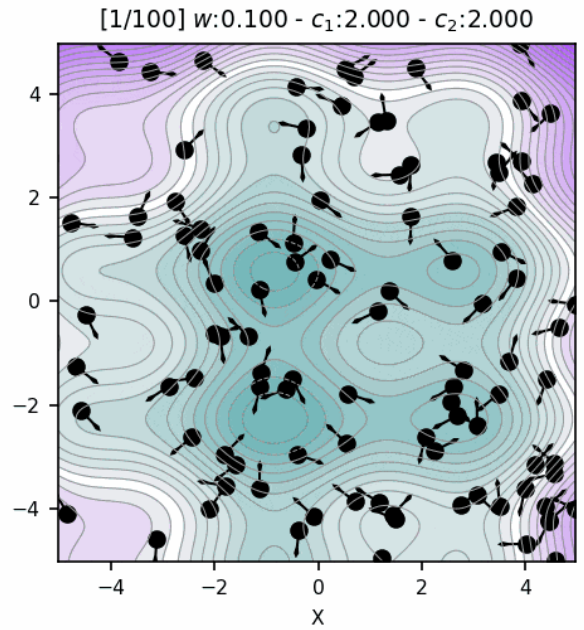


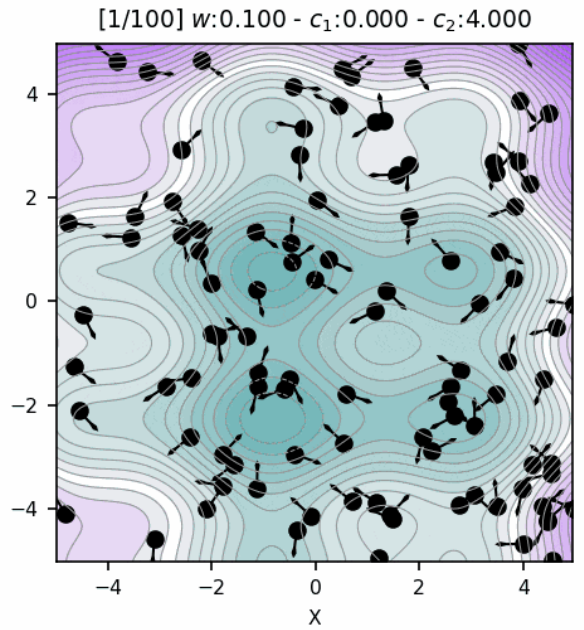


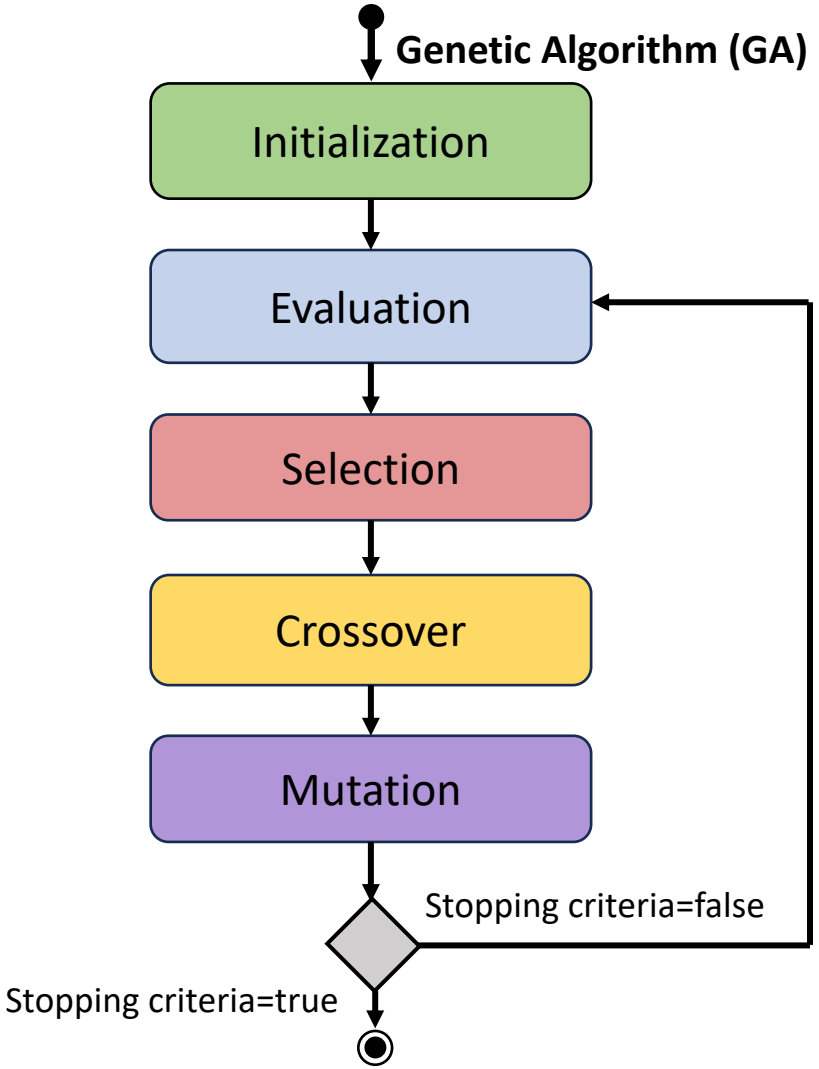


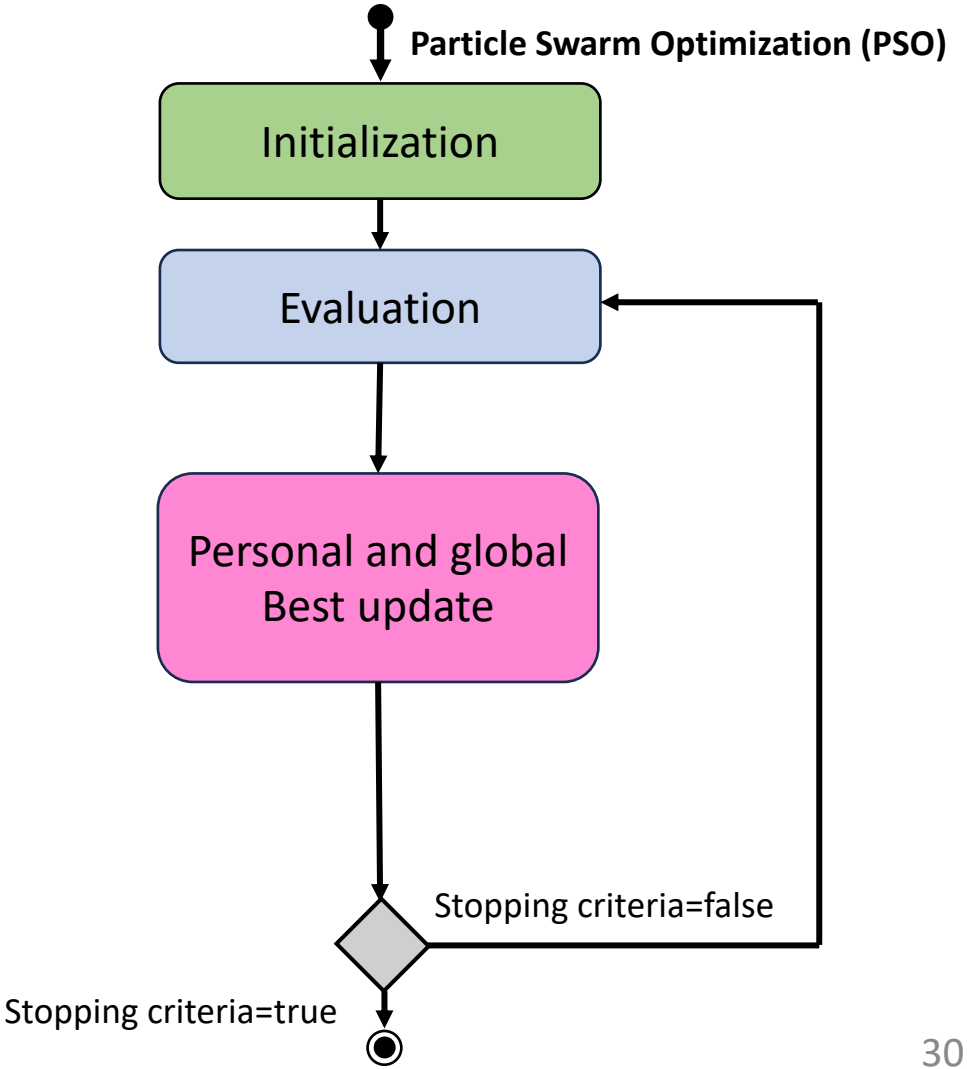
Particle Swarm Optimization (PSO)



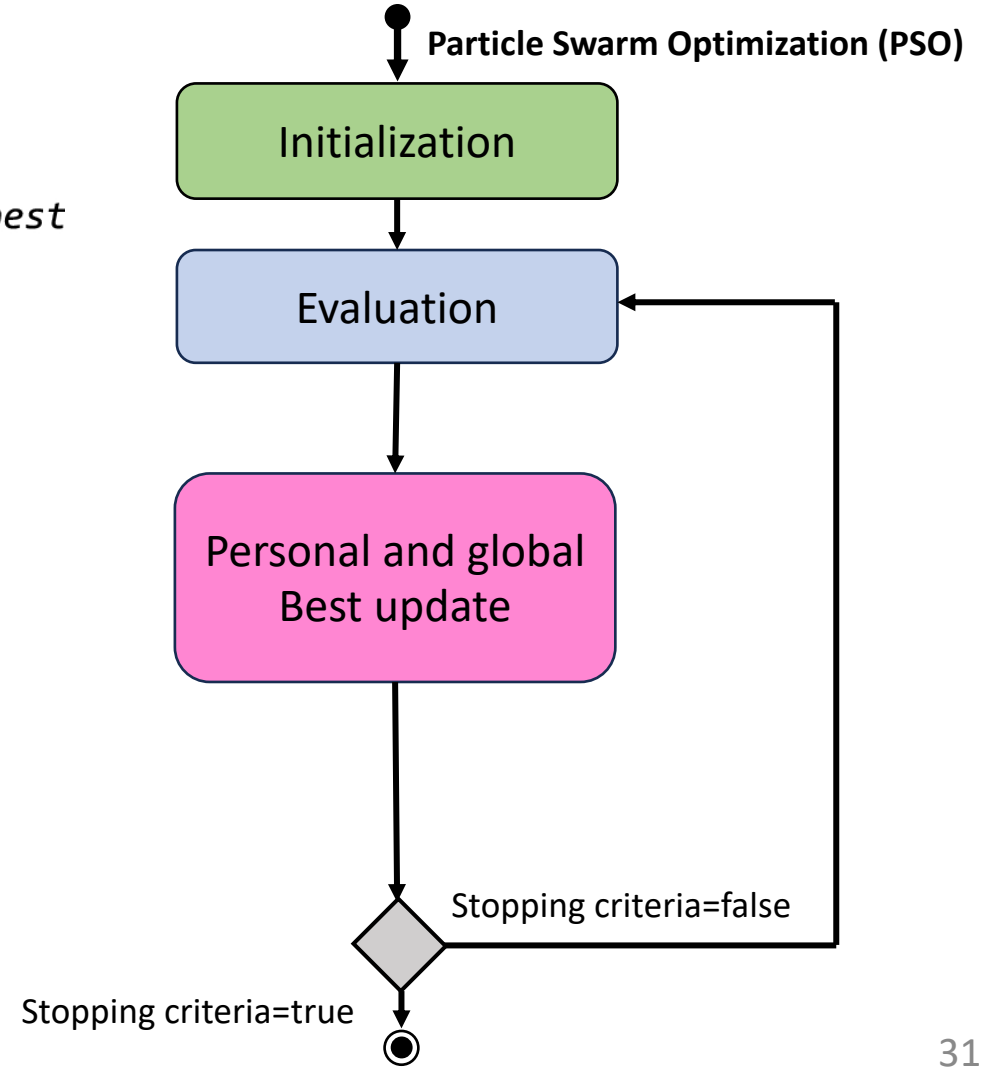




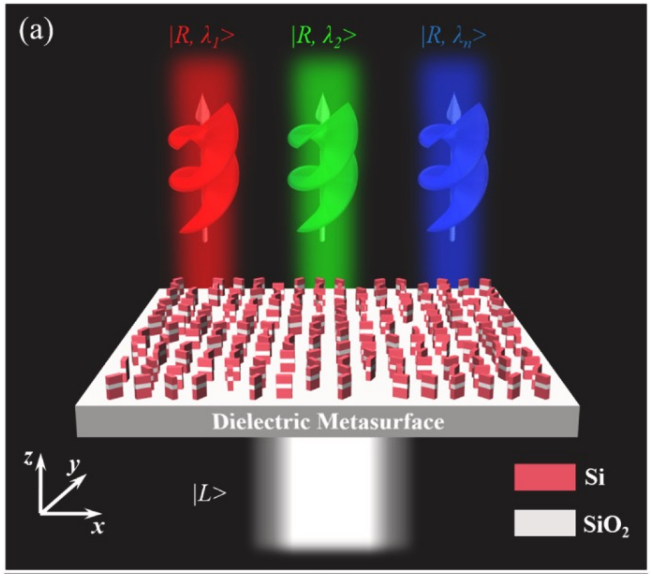

Genetic Algorithm (GA)
Initialization
Evaluation
Selection
Crossover
Mutation
Stopping criteria=false
Stopping criteria=true

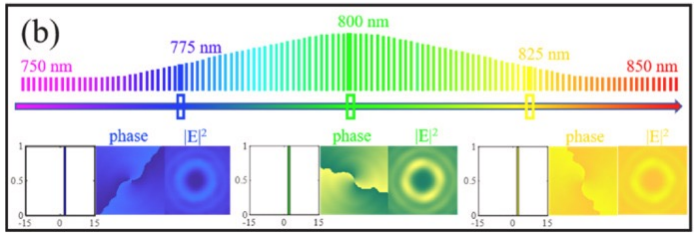

Particle Swarm Optimization (PSO)
Initialization
Evaluation
Personal and global
Best update
Stopping criteria=false
Stopping criteria=true

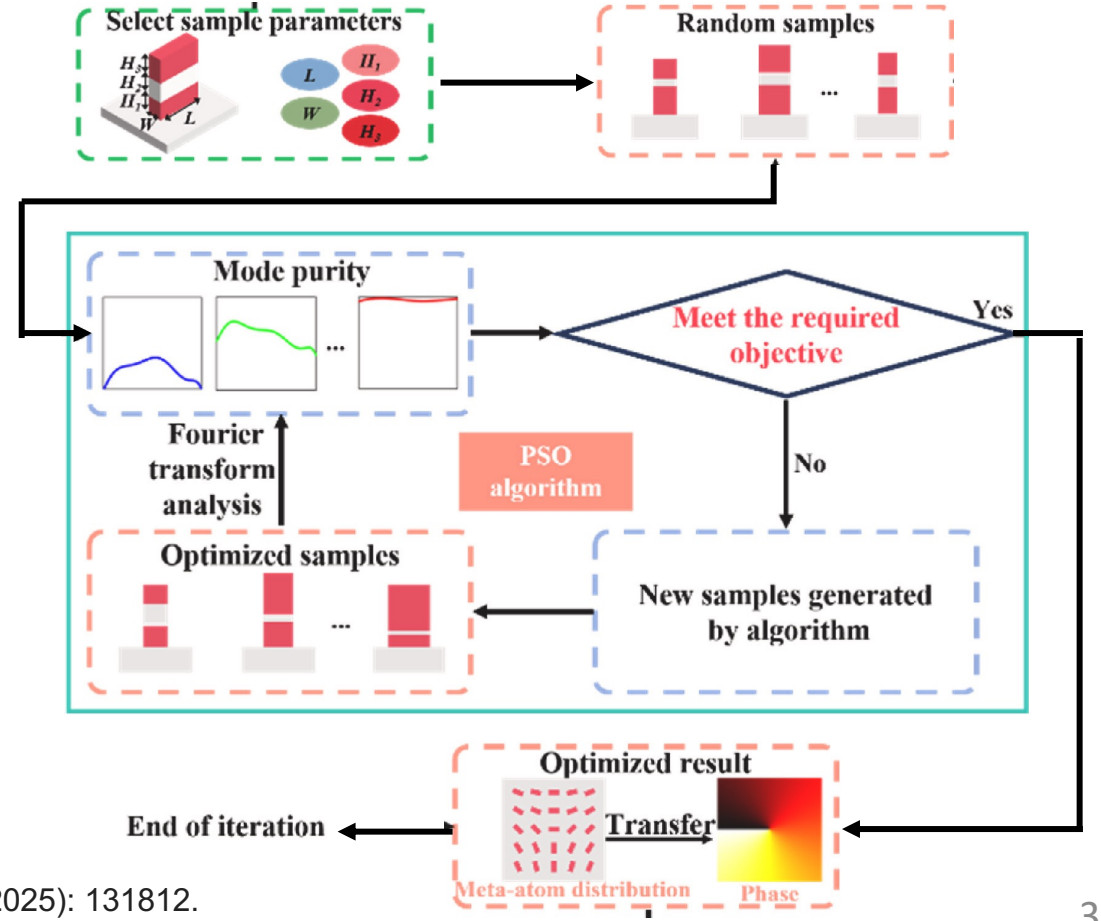
```
Initialize Xi, Vi, iteration, pbest, gbest
Generate random particles (P)
For each particle (i)
   Calculate fitness function (fi)
   Update pbest, gbest
End for
While iteration
   For each particle I
      Update Vi, Xi
      If Xi > limit then Xi = limit
      Calculate fitness function fi
      Update pbest, gbest
   End for
End while
```



Metasurface for generating high-purity vortex pulses

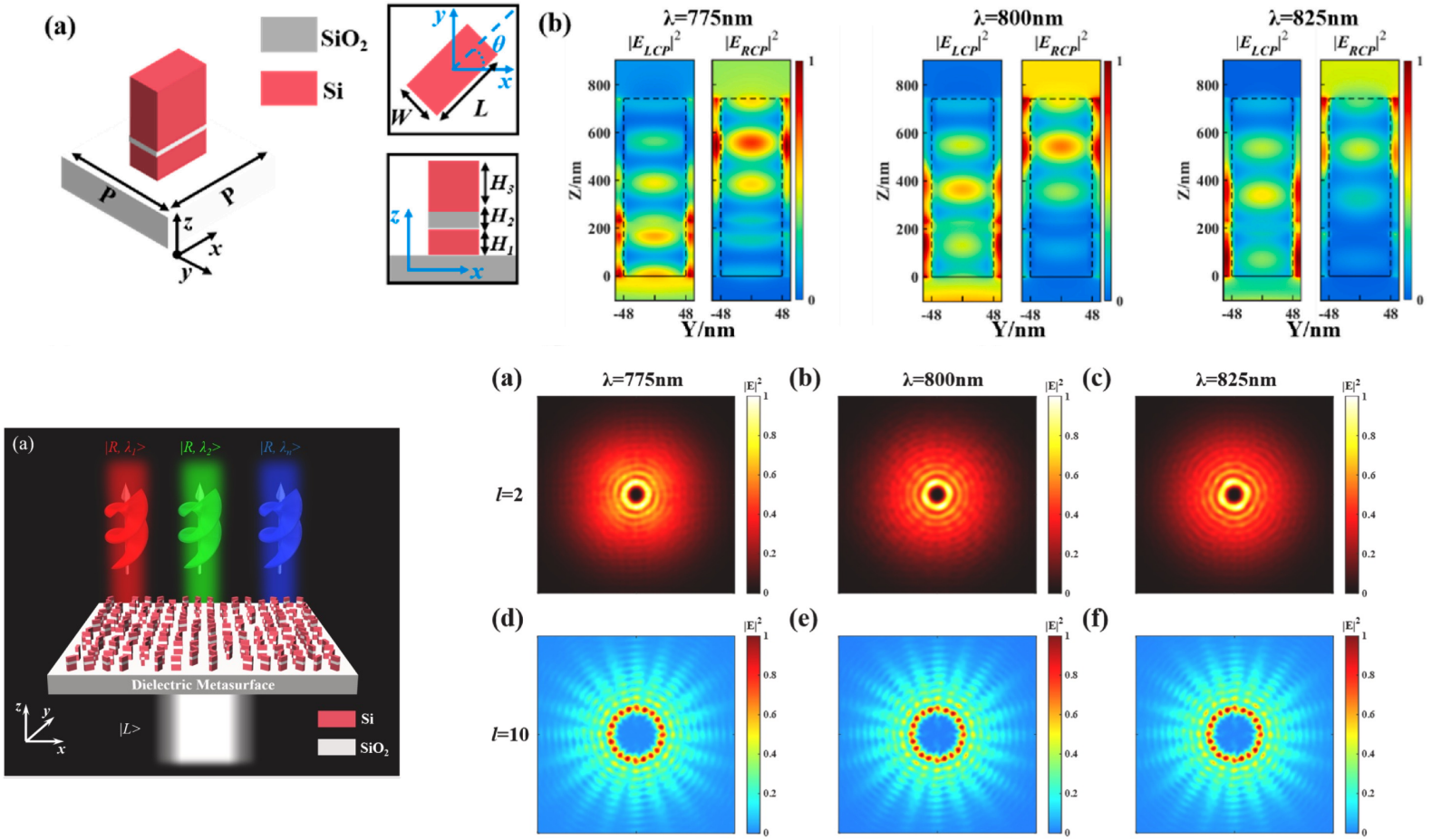




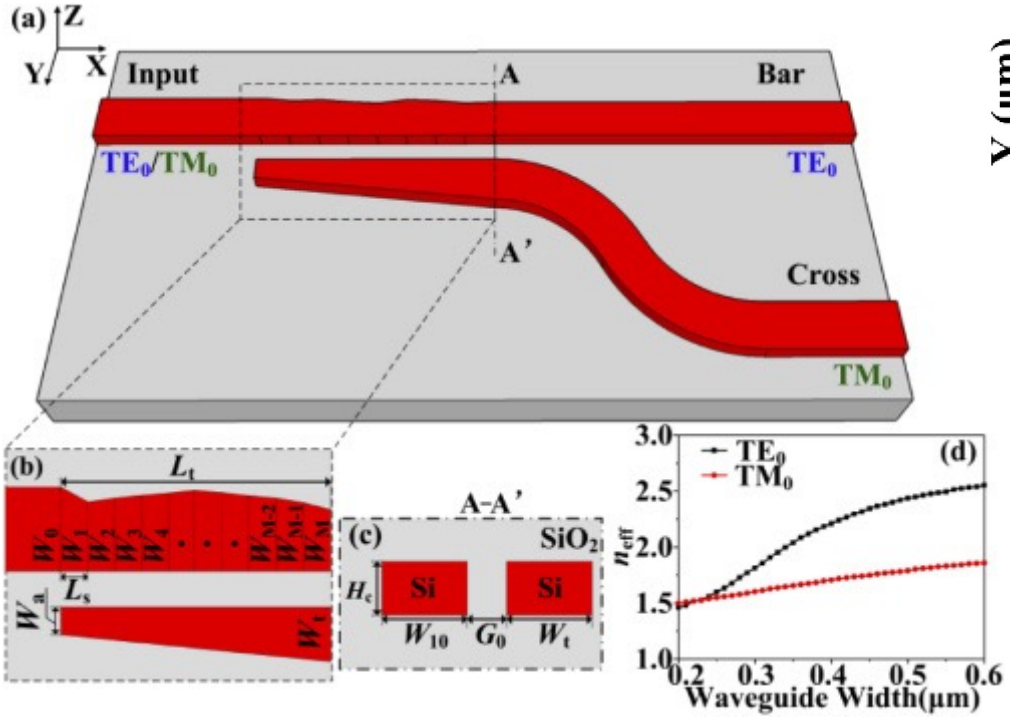


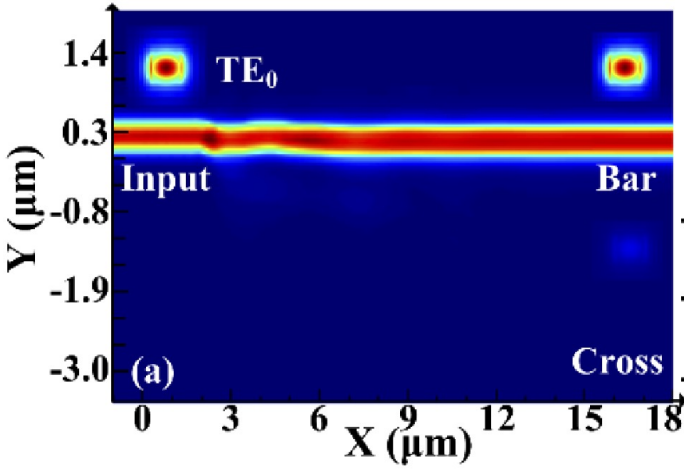
Xu, Jianghong, et al. *Optics Communications* 584 (2025): 131812.

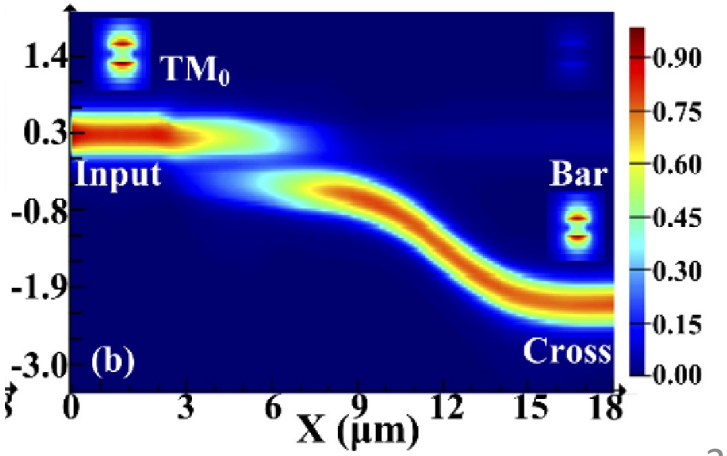
Metasurface for generating high-purity vortex pulses



silicon polarization beam splitter

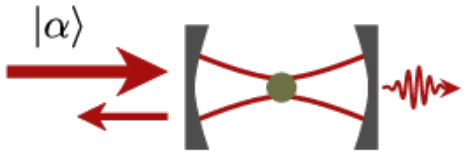


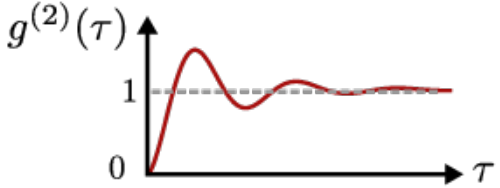




Optimizing dynamical blockade

Suppresses multiphoton generation in a bosonic mode;
generation of single photons through coherent optical driving





$$g^{(2)}(0) = 0, \quad g^{(2)}(\tau) \to 1$$

A single-photon source emits **one photon at a time**

$$X_i^k = \left[X_{i1}^k, X_{i2}^k, \cdots, X_{id}^k\right]^T$$

$$V_i^k = \left[V_{i1}^k, V_{i2}^k, \cdots, V_{id}^k\right]^T$$

$$X_i^{k+1} = X_i^k + V_i^{k+1},$$

$$V_i^{k+1} = wV_i^k + f_1 r_1 \left(P_i^k - X_i^k\right) + f_2 r_2 \left(G^k - X_i^k\right)$$

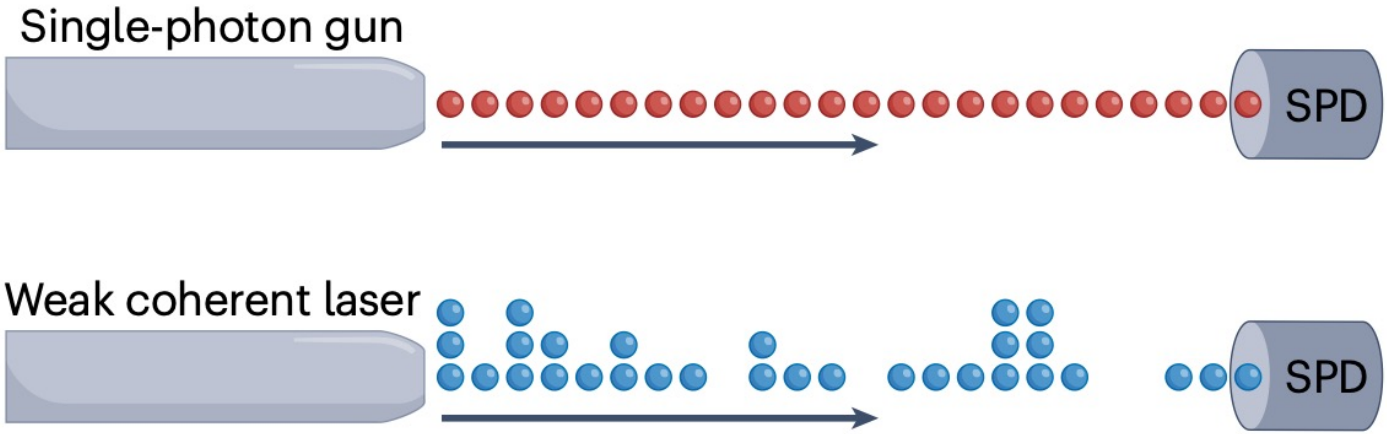
detuning driving strength

$$X_i^k = \left[\Delta_i^k, \varepsilon_{p,i}^k, T_i^k, A_i^k\right]^T$$

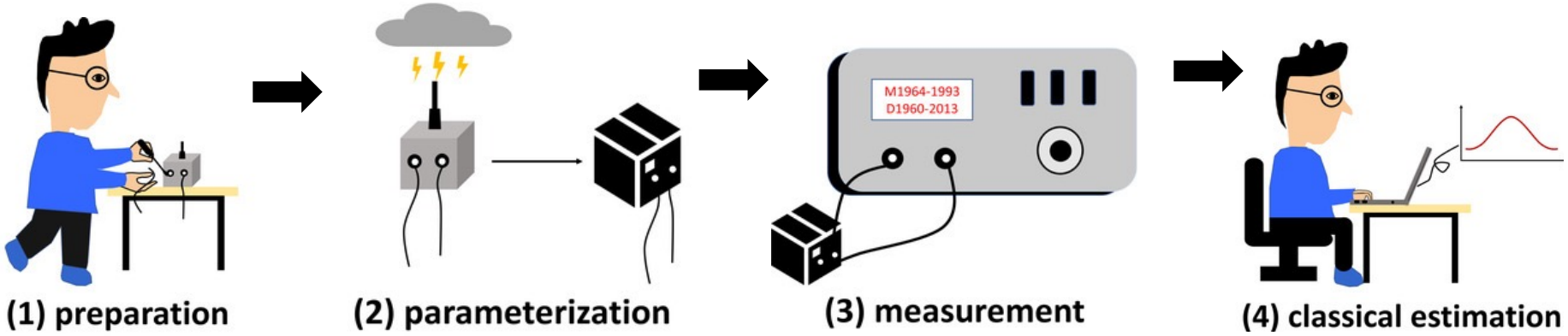
period the duration of the pulses

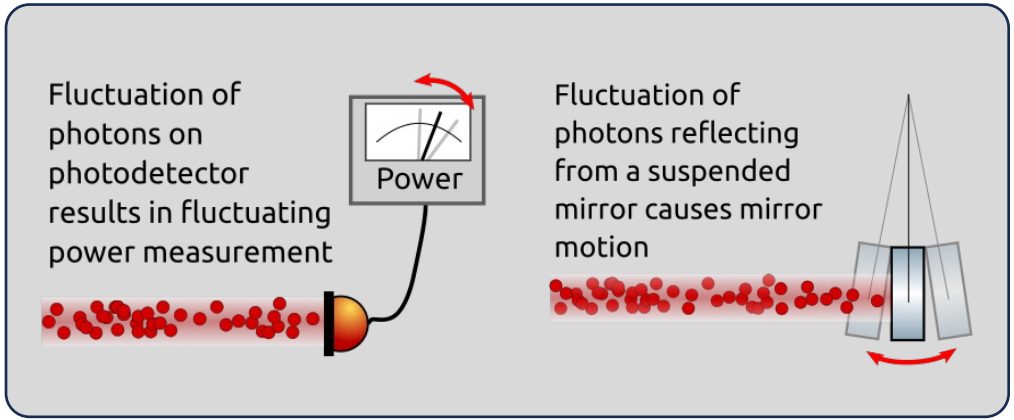
Zhang, Guang-Yu, Zhi-Hao Liu, and Xun-Wei Xu. “Optimizing dynamical blockade via a particle-swarm-optimization algorithm.” Physical Review A 110.2 (2024): 023718.

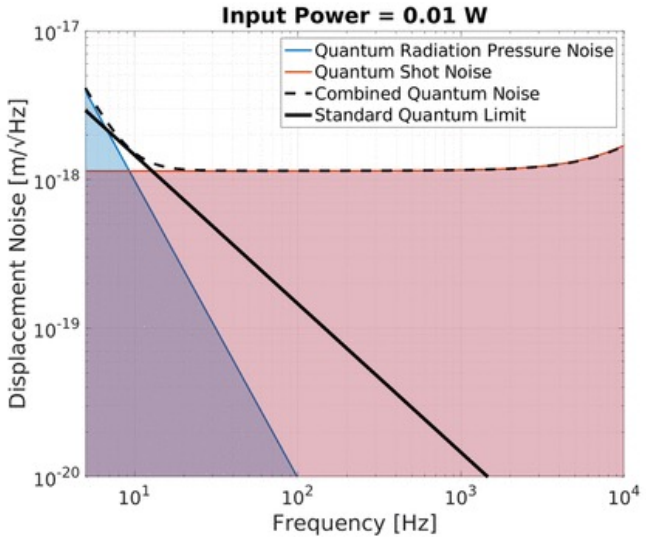
Optimizing dynamical blockade



Precise Quantum Measurement

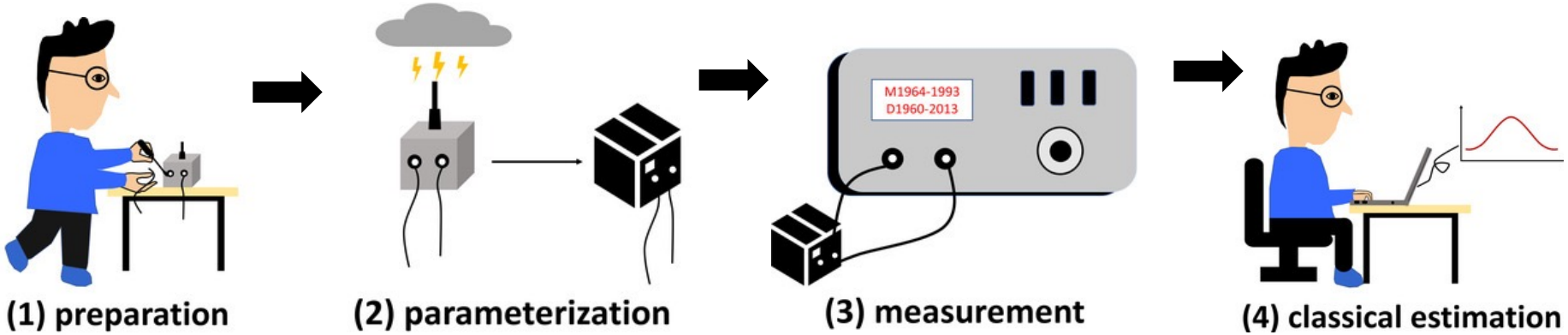


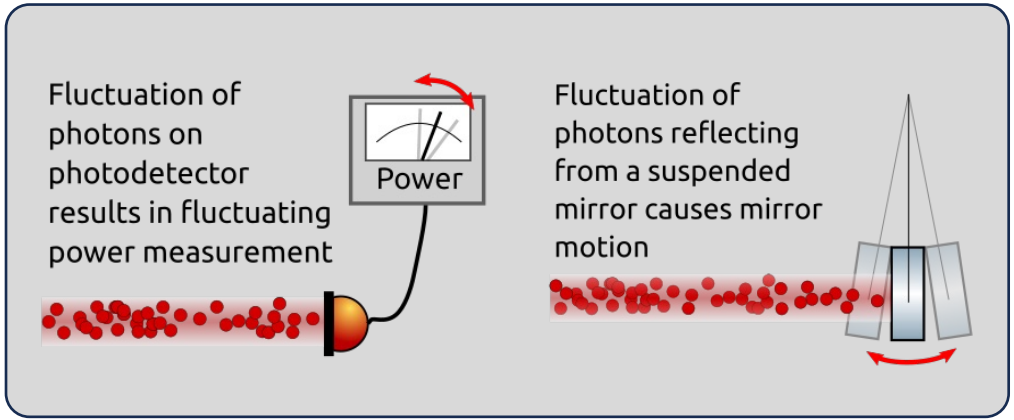




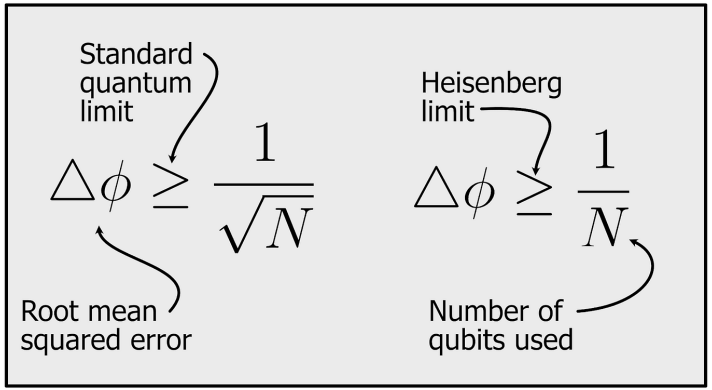
https://10m.aei.mpg.de/standard-quantum-limit-sql/

Precise Quantum Measurement





https://10m.aei.mpg.de/standard-quantum-limit-sql/

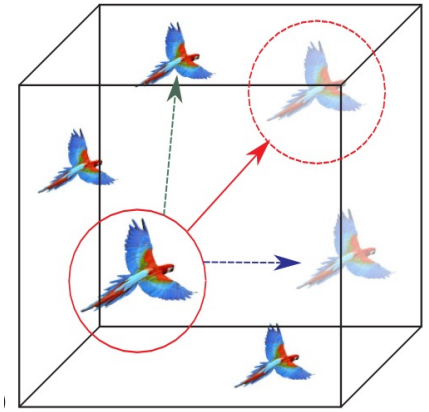


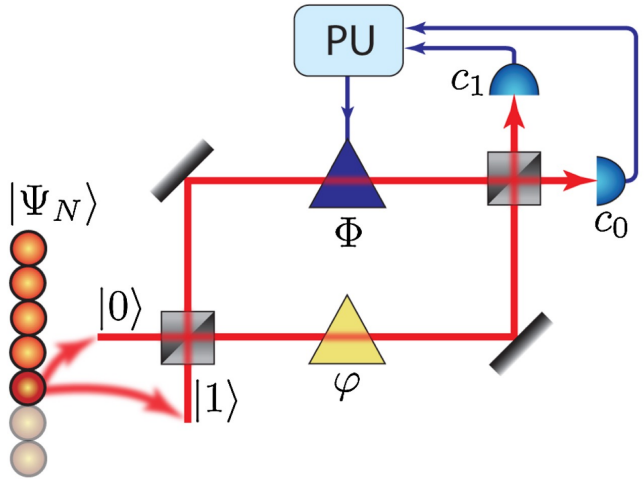
$$\Delta x \Delta p \geq \frac{\hbar}{2}$$

Precise Quantum Measurement

The goal is to send individual photons into the interferometer, and modify the controllable phase according to previous measurement results to decrease the phase uncertainty.

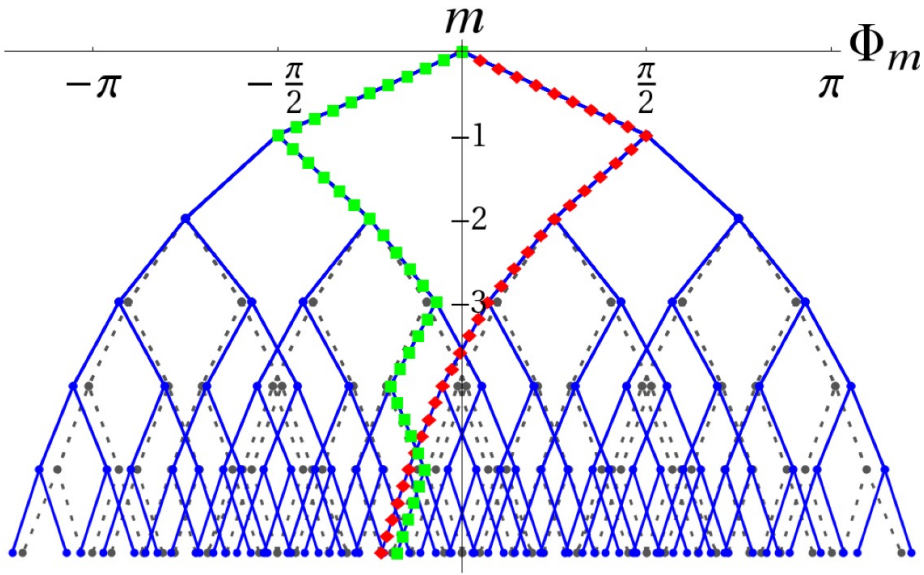
It can be formulated as a decision tree, where the actions must be chosen based on the measurement outcomes to maximize the information gain





$$p^i = (\Delta\Phi_1, \Delta\Phi_2, ..., \Delta\Phi_{N-1}, \Delta\psi)$$

$$\Phi_m = \Phi_{m-1} - (-1)^{u_m}\Delta\Phi_m$$



https://doi.org/10.1103/PhysRevLett.104.063603

Particle Swarm Optimization (PSO)

- Easy to understand and implement
- Very few algorithm parameters to adjust
- Derivative free
- Very efficient global search algorithm
- No evolution or mutation in the operator
- Easily parallelization for concurrent processing



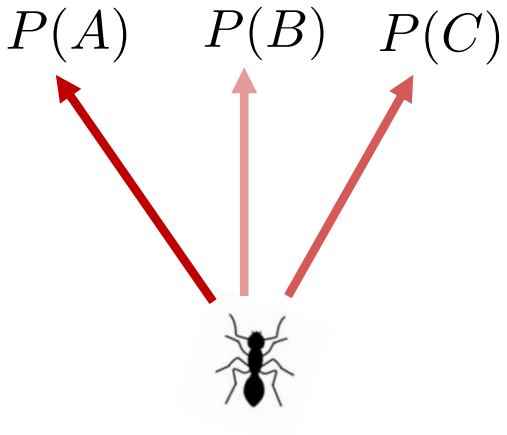
Slow convergence
Needs memory



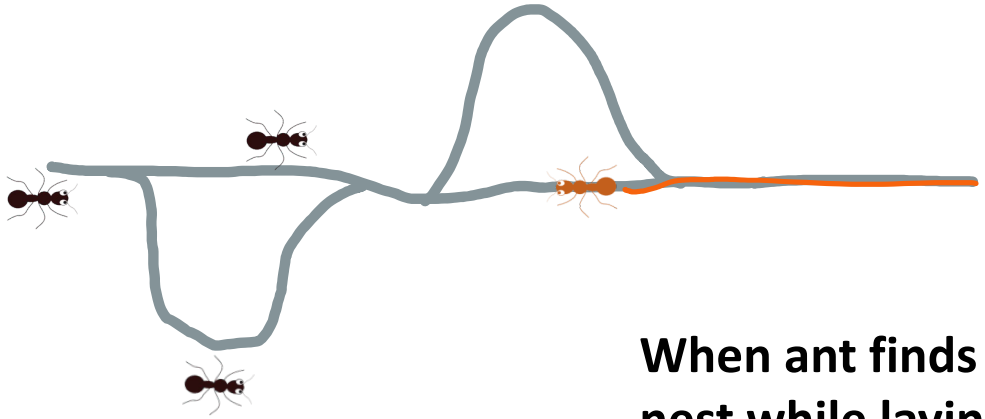
Ant Colony Optimization (ACO)

Ants deposit a certain amount of pheromone while walking, and each ant probabilistically prefers to follow a direction rich in **pheromone**.



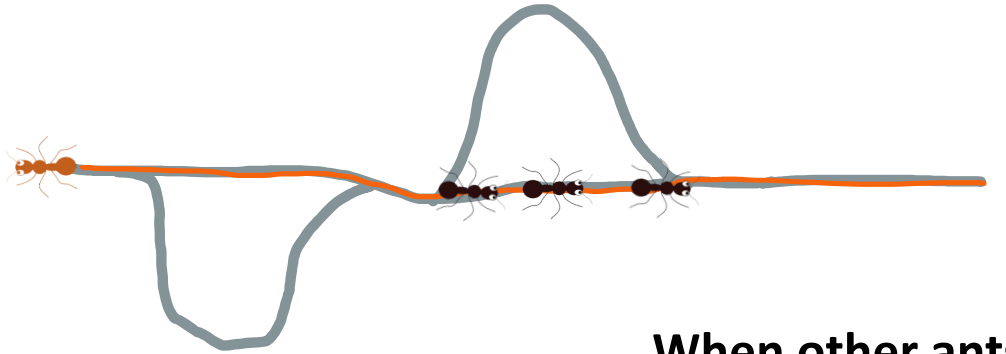


Ant Colony Optimization (ACO)



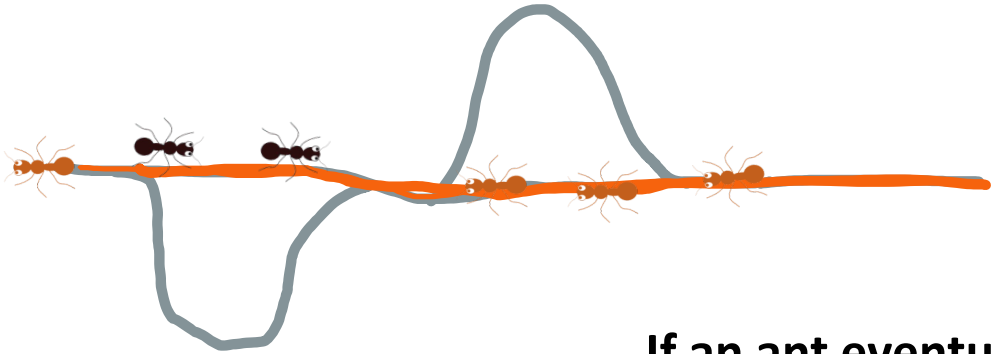
When ant finds food, it returns to the nest while laying down pheromones trail

Ant Colony Optimization (ACO)



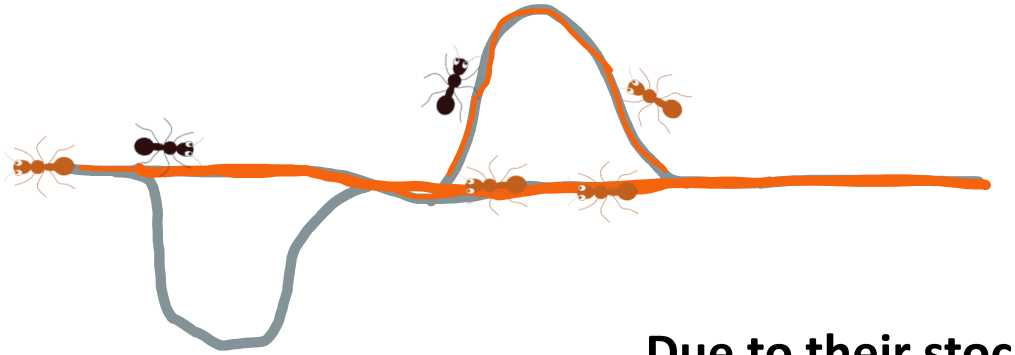
When other ants find a pheromone trail, they are likely not to keep travelling at random, but to instead follow the trail.

Ant Colony Optimization (ACO)



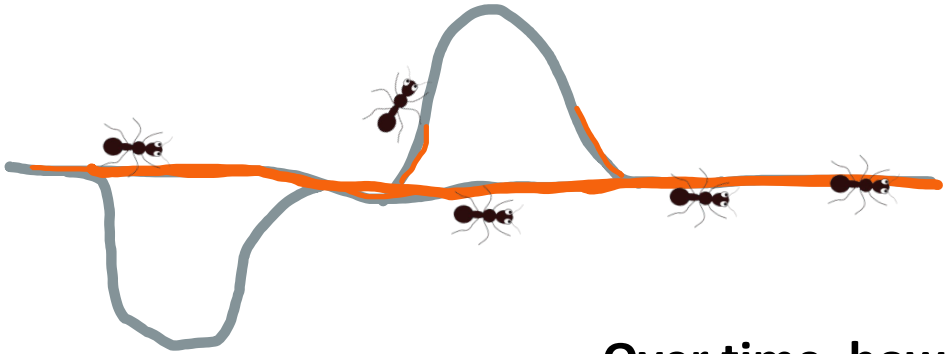
If an ant eventually find food by following a pheromone trail, it returning to the nest while reinforcing the trail with more pheromones.

Ant Colony Optimization (ACO)



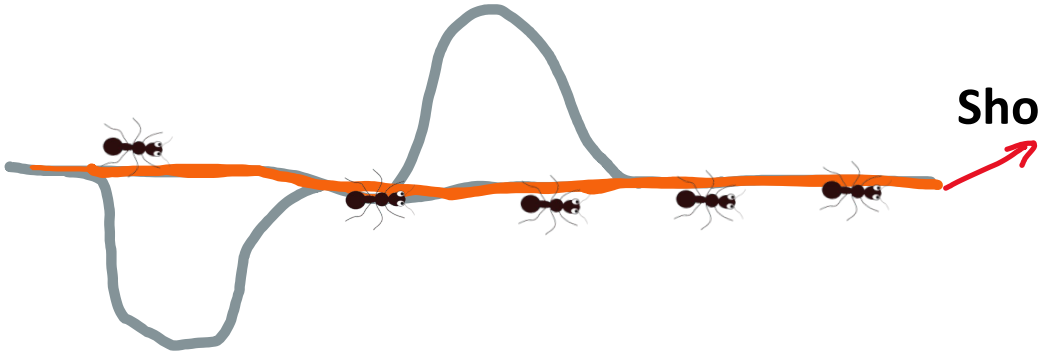
Due to their stochastic behaviour, some ants are not following the pheromone trails, and thus uncover more possible paths

Ant Colony Optimization (ACO)



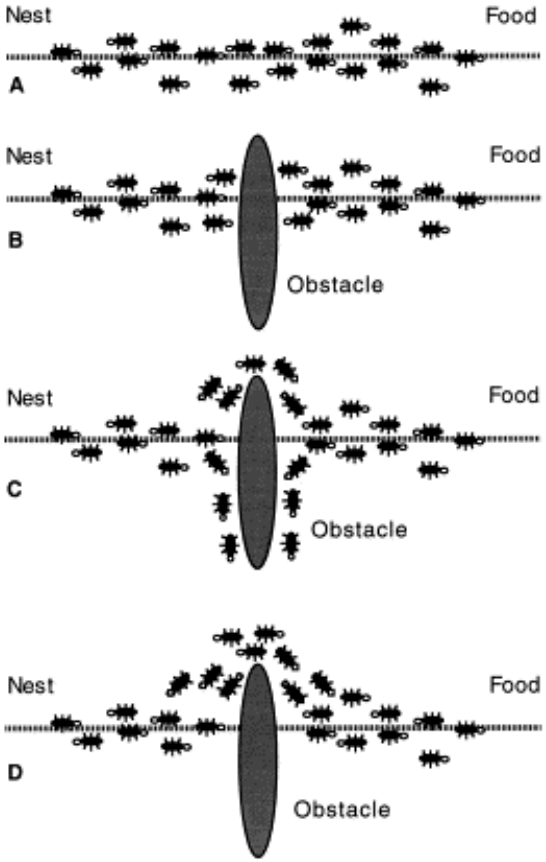
Over time, however, the pheromones trails start to evaporate, thus reducing its attractive strength

Ant Colony Optimization (ACO)



Shortest path is being obtained

Ant Colony Optimization (ACO)



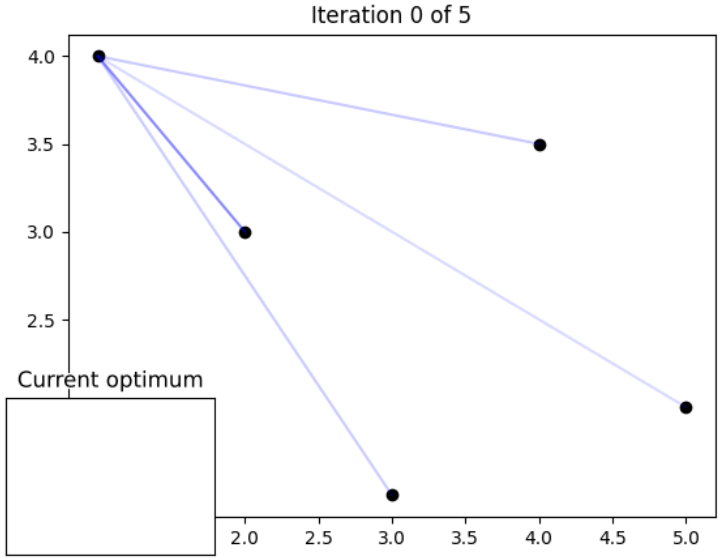
Ant Colony Optimization (ACO) is a probabilistic algorithm inspired by the behaviour of ants to find optimal paths through graphs.

solve discrete optimization problems

Originally applied to Travelling Salesman Problem

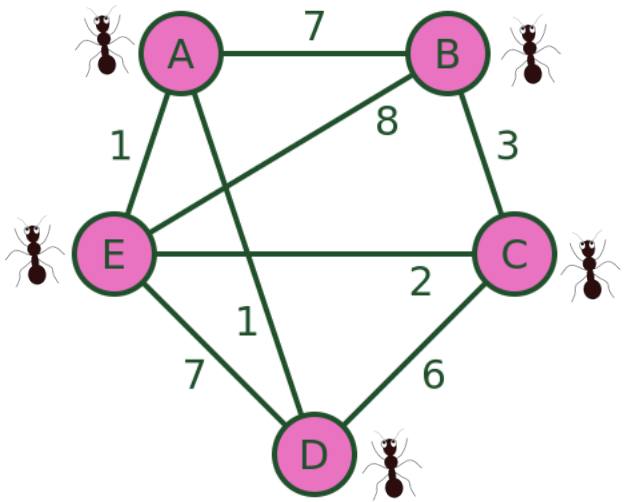


Marco Dorigo in 1992



Ant Colony Optimization (ACO)

Travelling salesman problem: Given a list of cities and the distances between each pair of cities, what is the shortest possible route that visits each city exactly once and returns to the origin city

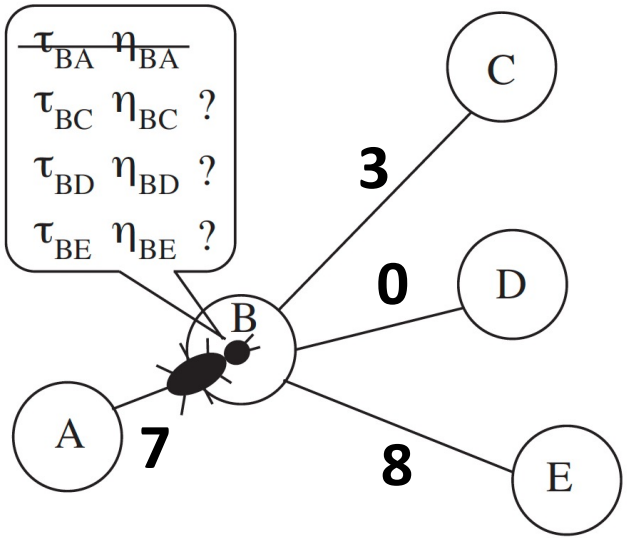


1. Place ants at different nodes
2. Find the path for each ant
3. Update pheromone intensity
4. Repeat 2-3 until stopping criteria have been satisfied

```
procedure ACO_MetaHeuristic is
  while not terminated do
    generateSolutions()
    daemonActions()
    pheromoneUpdate()
  repeat
end procedure
```

Ant Colony Optimization (ACO)

A virtual ant arriving from node A considers which edge to choose next on the basis of pheromone levels τ_{ij} and visibilities η_{ij} (inverse of distance). The edge to node A is not considered because that node has already been visited



Probability that ant k chooses the edge from city i to city j

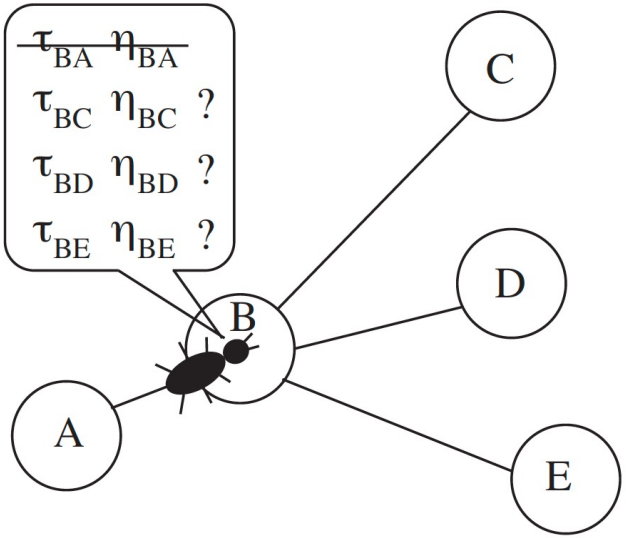
$$p_{ij}^{k} = \frac{\tau_{ij}^{a}\eta_{ij}^{b}}{\sum_{h\in J^{k}}^{H}\tau_{ih}^{a}\eta_{ih}^{b}}$$

pheromones

the inverse of the edge length

If a = 0, ants choose solely on the basis of the shortest distance;
if b = 0, ants choose solely on the basis of the pheromone amount

Ant Colony Optimization (ACO)



$$p_{ij}^{k} = \frac{\tau_{ij}^{a}\eta_{ij}^{b}}{\sum_{h \in J^{k}}^{H} \tau_{ih}^{a}\eta_{ih}^{b}}$$

Once all ants have completed a tour of the graph, each ant k retraces its own path and deposits an amount of pheromone on travelled edges according to

an amount of pheromone — constant — the total length of the path found by ant k

$$\Delta\tau_{ij}^{k} = Q/L^{k}$$

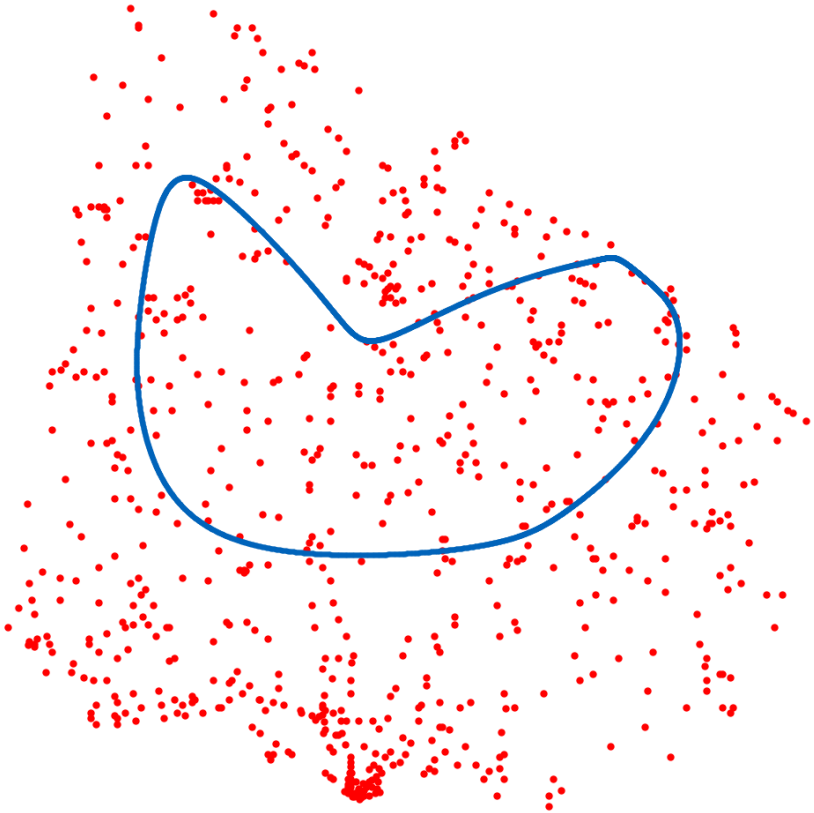
The amount of pheromones on each edge after all M ants have retraced their own path is equal to

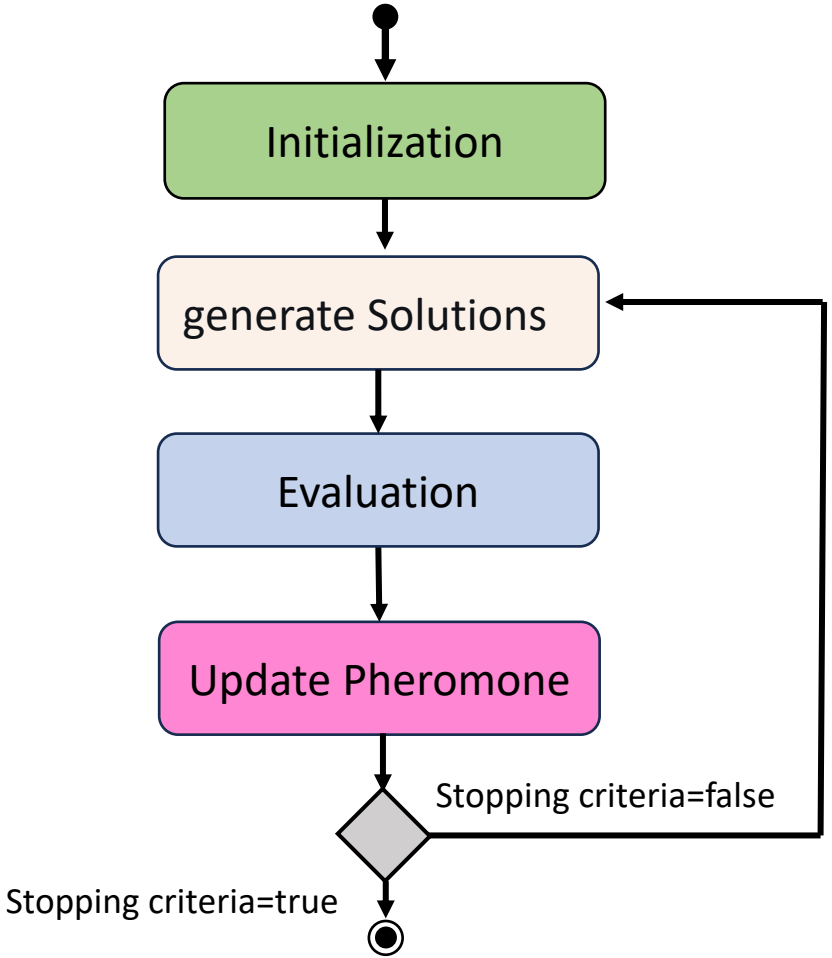
$$\Delta T_{ij} = \sum_{k}^{M} \Delta\tau_{ij}^{k}.$$

$$\tau_{ij}^{t+1} = (1-\rho)\tau_{ij}^{t}\Delta T_{ij}$$

pheromone evaporation

Ant Colony Optimization (ACO)





Ant Colony Optimization (ACO)

- Works extremely well for discrete / combinatorial problems (TSP, routing, scheduling).
- balance between exploration (evaporation) and exploitation (pheromone reinforcement)
- Derivative free
- No evolution or mutation in the operator
- Naturally models path-based structures, making solutions interpretable (pheromone levels show learned structure)



- Could be computationally expensive (many ants × many iterations)
- Highly sensitive to parameter tuning
- Risk of premature convergence when pheromone concentrates too quickly
- Poor scalability to very large problems
- More complex to implement than PSO
- Less effective on continuous optimization problems



Discussion

What is Creativity?


Colorado State Fair's annual art competition

What is Creativity?

Creativity in Art vs Creativity in Science&Tech?

Creative Artist or Creative Artwork?

How to detect creativity?


Colorado State Fair's annual art competition

Can animals be creative?

Can computers be creative?

What is Creativity?

Creative Process vs. Creative Results

Poincare: "*Preparation, incubation Insight, revision*"

From Cognitive Science:
"*Problem Definition, Idea Generation Idea Evaluation, Idea Selection*"

"Standard Definition of Creativity"
[Runco,Jaeger 2012]:
Originality & Effectiveness

Originality = Novelty

Effectiveness: Usefulness, utility

Originality is vital, but must be balanced with fit and appropriateness.